Tactical choice in repeated protests

Elaine Yao*

October 16, 2026

Abstract

Why do some protest movements switch between peaceful and violent tactics, while others remain committed to one approach? I argue tactical choice is central to how movements manage declines in participation driven by past failures. I develop a dynamic model in which a movement leader chooses between tactics that differ in how they convert participation into political power. Failed protests reveal information about regime strength, reducing subsequent mobilization. Optimal tactical sequencing reflects not only how powerful each tactic is today, but how valuable it may be in a lower-mobilization environment tomorrow. I show that organizers may begin peacefully even when violence offers greater immediate prospects of success, preserving tactics that remain effective after demobilization. I also show that sufficiently severe failures favor peaceful continuations. An extension shows that repression aimed at reducing tactical effectiveness can change the informational content of failures, partly or even fully undermining its own effectiveness.

*Assistant Professor, Washington University in St. Louis. E-mail: eyao@wustl.edu

I am grateful to Germán Gieczewski and Matías Iaryczower for their guidance throughout the development of this project. I am also grateful for the very useful suggestions from Gaétan Nandong, Scott Gehlbach, Carlo Prato, Daniel Goldstein, Keith Schnakenberg, Carlo Horz, Juan Dodyk, Francesco Giovannoni, Carly Wayne, and numerous seminar participants at Columbia, WashU, SITE, National Taiwan University, the PoleconUK webinar, MPSA, and APSA. All remaining errors are fully my own.

1 Introduction

What determines how protest movements choose and change tactics over time? Some movements undergo striking transformations in their main mode of contestation: the African National Congress shifted from civil disobedience to armed sabotage and insurgency in its fight against South African apartheid,[1] while the Zapatista movement in Mexico moved in the opposite direction, developing into a broad-based and largely nonviolent movement after its armed 1994 uprising (Inclán 2018). Other movements maintain a largely consistent tactical approach even through severe setbacks: Poland's Solidarity movement remained nonviolent through martial law and years of repression, while Algeria's National Liberation Front waged armed struggle for independence from the outset. What explains these divergent trajectories, and how do movement leaders decide whether to persist with or depart from their initial tactics?

Canonical theoretical accounts emphasize how nascent protest movements grow by exposing latent sympathies, lowering barriers to participation, and generating mobilization cascades (Kuran 1991; Lohmann 1994). Yet the same informational feedback loop that helps launch a protest can work in reverse as failed episodes accumulate. When a movement repeatedly falls short of its objectives, citizens gather evidence that the regime is stronger and the movement's prospects worse than they previously believed. Failure therefore erodes confidence in subsequent attempts and, with it, the base of participation on which many forms of contention depend. Thus, while existing accounts tell us a great deal about how information makes movements grow or peter out, a key open question is how movements change what they do as public confidence deteriorates.

This paper argues that tactical choice is central to how movements manage declining public confidence. The key is that different tactics convert participation into political leverage in different ways. Mass-based tactics such as marches, occupations, and popular uprisings generate leverage through numbers: by demonstrating the breadth of opposition and paralyzing the normal functioning of society, they can compel policy concessions or regime change. By contrast, disruptive tactics such as sabotage or insurgency exert coercive pressure with relatively few participants, relying on resources or arms to damage infrastructure, disrupt economic activity, or threaten state assets. Mass-based tactics are therefore most powerful when participation is broad but lose force as turnout falls, whereas disruptive tactics gain less from numbers but hold up better as support erodes.

[1] While on trial, Nelson Mandela justified the tactical switch, arguing that the ANC's reliance on nonviolent resistance had "achieved nothing," and supporters were "beginning to lose confidence in this policy." He defended sabotage as a means of exerting pressure through, among other things, "attacks on the economic life-lines of the country" (Mandela 1964).

Because movement leaders strategically anticipate that failure will depress turnout, the most powerful tactic today need not be the one they deploy first. Leaders weigh not only a tactic's immediate force but also the damage its failure would do to public beliefs: the more powerful a protest that fails, the stronger the evidence it provides of regime strength, and the more sharply it discourages future participation. Tactical choice thus allows leaders to shape how far mobilization declines – and then to adapt to the declines that do occur. They sequence tactics accordingly, capitalizing on expected turnout today while positioning the movement for subsequent attempts.

I formalize these ideas in a two-period model in which an organizer chooses between two stylized protest technologies. *Peace* represents a participation-intensive technology, generating little leverage when turnout is low but scaling strongly with participation. *Violence*, by contrast, generates substantial leverage even with limited participation, but benefits less from additional turnout. These labels distinguish tactics by how their effectiveness varies with participation, rather than by the literal presence or absence of violence. A mass revolution may be better captured by the former technology if its effectiveness depends primarily on the mobilization of large numbers, even if it involves armed violence. After the organizer chooses one of the two tactics, citizens decide whether to participate, and protest succeeds if the power generated exceeds the unknown strength of the regime. If protest in the first period fails, citizens infer that the regime is stronger than previously believed, reducing their willingness to participate, and the game is played again in period 2.

My first set of results shows that even when the violent tactic is extremely effective, it may not be deployed immediately – or even at all. In fact, organizers may deliberately preserve violence for later, and following a sufficiently *severe* failure, continue to use peace in the second period. First, because tactics differ in their sensitivity to participation, their relative effectiveness changes as mobilization declines. Violent tactics remain effective in pessimistic, low-mobilization environments; by contrast, participation-intensive "people power" tactics are *perishable*: their effectiveness depends on broad mobilization, which can dissipate after setbacks. This creates a "use-it-or-lose-it" logic: even when violence offers greater *immediate* leverage, an organizer may begin peacefully, preserving violence as insurance against future demobilization. This incentive is strongest when turnout is particularly sensitive to setbacks: the more sharply failures reduce participation, the more valuable it becomes to deploy participation-intensive tactics early, when they are likeliest to be most effective. Thus, a movement need not deploy its highest-leverage tactic immediately.

The second reason is that larger first-period failures make the peaceful tactic more appealing in the second period. Failure has two opposing effects on subsequent tactical choice. It

reduces expected turnout, increasing the appeal of violent tactics, which can generate pressure with few participants. However, it also provides evidence that the regime is strong, raising the amount of leverage required for any subsequent challenge to succeed. After smaller failures, the first force dominates. However, after sufficiently severe failures, the second force outweighs it: Violence may remain relatively effective at low turnout, yet still lack the upside needed to overcome a strong regime. The organizer therefore gambles on peace, whose higher ceiling offers better prospects against the stronger regimes that remain. The result is a *selection-based* account of why not all failed protests result in subsequent violence: the failures that are most informative about regime strength may also make participation-intensive, high-upside tactics comparatively appealing for the future.

Existing explanations typically attribute different movements' tactical trajectories to distinct strategic logics, such as the radicalization of remaining participants (della Porta 1995; McCauley and Moskalenko 2008), the exacerbation of grievances by violent state repression (Ritter and Conrad 2016), or systematic differences in the costs of participation under different tactics (Chenoweth and Stephan 2011). While these mechanisms undoubtedly matter in many settings, I show that they are not necessary to generate diverse patterns of tactical change. Depending on their environment, movements in the model may remain peaceful throughout, begin peacefully and turn to violence after weak failures, or begin violently and turn to peace after sufficiently powerful failures. These trajectories broadly resemble those of Solidarity, the ANC, and the Zapatista movement, respectively.

To make these predictions concrete, I interpret the model to explain how the political and physical environment shapes tactical sequencing. Urbanization and urban redevelopment, for example, can reduce the leverage available to small groups while making street fighting, occupations, or control of public space more dependent on large crowds, weakening violence as a post-demobilization fallback even while strengthening its appeal during periods of broad mobilization.

In an extension, I study how repression interacts with the baseline dynamics. Policing, restrictions on public space, and investments in security alter the tactical environment, but their efficacy is limited by two factors. The first is simple tactical substitution: weakening one tactic can redirect contention rather than suppress it. Repression can therefore induce tactical switches opposite to those prompted by learning alone in the baseline model.

The second limit is informational. Repression may help the regime survive an initial protest, but it also changes what citizens learn from failure. Failure under little repression provides strong evidence that the regime itself is difficult to defeat; failure under heavy repression can

instead be attributed partly to the intervention, leaving citizens less pessimistic. The regime thus faces a trade-off between the immediate protection afforded by repression, and its effect on future mobilization. When mobilization is highly responsive to beliefs, repression can be self-defeating: by making failure less discouraging, it preserves optimism and future turnout, leaving the movement better positioned to mobilize again.

The paper proceeds as follows. Section 2 discusses related literature, and Section 3 presents the baseline model. Section 4 abstracts away from tactical choice to isolate the relationship between learning from failures and mobilization. Section 5 analyzes tactical sequencing and how structural conditions shape movement trajectories. Section 6 extends the model to repression, and Section 7 concludes.

2 Literature

The paper builds on the classic literature on repertoires of contention, which emphasizes that movements draw upon and innovate upon historically conditioned sets of contentious tactics in response to contemporary circumstances (McAdam 1983; Tilly 1978). My contribution is to examine the strategic logic of *sequencing* tactics within a given repertoire. Rather than explaining how new tactics emerge or become available, I study how organizers choose among existing tactics as previous episodes of contention change the conditions under which they can be effectively deployed.

The model is also related to work on how rebel groups choose among different tactics or technologies of contention. In particular, Bueno de Mesquita (2013) takes a very similar approach to modeling tactics: he treats conventional and irregular tactics as technologies that differ in how effectively they convert mobilization into political power. In his model, organizational capacity is exogenous and tactical choice follows participation; here, mobilization is endogenously pinned down by beliefs and learning. Qiu (2022) likewise studies dynamic tactical choice, and shows that guerrilla warfare allows rebels to survive and accumulate strength, creating incentives to postpone conventional confrontation until conditions improve. Here, failed protests can undermine future mobilization, creating incentives to deploy participation-intensive tactics early. Morris and Shadmehr (2024) study the choice of repertoire *width*, but focus on which set of contentious actions is made available in a one-shot environment rather than on tactical sequencing across repeated attempts.

A thematically related study of dynamic tactical choice in protest is developed by Boyer et al. (2026). Their model articulates the idea that tactical choice may generate learning about the *ideological composition* of the movement. For instance, online mobilization initially draws

in large numbers of participants, but also facilitates the emergence of radical factions whose use of violence subsequently crowds out moderates. As such, this explanation complements work that focuses on movement "condensation," whereby remaining protesters undeterred by state repression tend to be more radical (McCauley and Moskalenko 2008), as well as a separate stream of theoretical work that emphasizes how organizations may become "captured" by their most radical members (Gieczewski 2021; Gieczewski and Kosterina 2024).

My model contributes directly to a growing theoretical literature on dynamic coordination under incomplete information (Dasgupta 2007; Dasgupta, Steiner and Stewart 2012; Gieczewski and Koçak 2024; Mathevet and Steiner 2013), and the broader literature that uses the global games approach originated by Morris and Shin (1998, 2003) to study protests (Boix and Svolik 2013; Casper and Tyson 2014; Little, Tucker and LaGatta 2015; Shadmehr 2021; Shadmehr and Bernhardt 2017). The most direct antecedent is Angeletos, Hellwig and Pavan (2007) who model dynamic global games where regime survival conveys public bad news about regime strength. Little (2017) studies a closely related mechanism in the context of coups. Importantly, AHP show that public learning from regime survival can undo the uniqueness properties generated by static global games (see also Shadmehr (2019).) My technical contribution is to move private noise to participation costs; this, combined with support conditions that preserve two-sided dominance after every public history, restores uniqueness in a repeated global game even with public learning over regime strength.

Finally, the paper speaks to a broad literature on how tactical choices interact with movement composition, organizational resources, and structural conditions, and on the consequences of repression and violence by both protesters and the state. My results on how repertoires of contention shape movement trajectories connect to work on how the endowment and distribution of resources affect the forms of contention movements adopt (Blair 2024; Dube and Vargas 2013; Sonin and Wright 2024). The feedback in my model between tactics, learning, and mobilization also relates to research on how tactical choices shape participation and support, though this work emphasizes different mechanisms. Nonviolent resistance, for instance, may lower barriers to participation (Chenoweth and Stephan 2011). I further address this point in the discussion of model assumptions (section 3.1.) In short, appropriate cost differences can reproduce the individual participation decisions which I generate by differences in tactical effectiveness, but they may not also reproduce the dynamic implications central to this paper.

Other work emphasizes how violence can discourage subsequent participation or shift how outside audiences respond to a movement (Manekin and Mitts 2022; Simpson, Willer and Feinberg 2018; Steinert-Threlkeld, Chan and Joo 2022; Wasow 2020). Citizens in my model draw the same inferences regardless of which tactic is used, but the tactic production functions can

be read as reduced forms that already incorporate such effects: a lower return to participation under violent tactics may partly reflect the support those tactics cost the movement. In general, the aim of this paper is thus not to deny these mechanisms, but to isolate a distinct and under-theorized one: how the prospect of demobilization after setbacks shapes the sequencing of tactics, even when citizens' underlying preferences, the distribution of participation costs, and the available repertoire remain unchanged.

3 Model

Overview. There are two periods, $t \in \{1, 2\}$, and two sets of strategic actors: a protest organizer and a unit mass of potential participants. The organizer chooses the tactic used in each period in order to maximize the probability that the movement succeeds over its lifetime. Whenever the organizer is indifferent between the two tactics, he chooses peace.[2] The organizer is forward-looking and does not discount the future. The organizer also cannot compel participation: after observing the organizer's choice of tactic, citizens make individually optimal participation decisions.

The movement confronts a regime of unknown strength θ. Before the game begins, θ is drawn Unif$[0, M]$, with $M > 1$. Regime strength is fixed across periods. Let F_t denote the commonly held beliefs at the beginning of period t, so $F_1 = \text{Unif}[0, M]$. A successful protest ends the game; following a failed period 1 protest, the movement may try again in period 2.

Tactics and protest power. In each period in which the game remains active, the organizer chooses between two tactics: *peaceful* and *violent*, denoted by $\tau \in \{p, v\}$. Each tactic is associated with a continuously differentiable and weakly increasing production function $A_\tau(\ell)$, which maps turnout $\ell \in [0, 1]$ – the share of potential participants who join the protest – into protest power. A protest using tactic τ succeeds if and only if $A_\tau(\ell) > \theta$. For all $\ell \in [0, 1]$, assume:

(A1) $A_v(0) > A_p(0) = 0$: violence has higher baseline power;

(A2) $A_v(1) < A_p(1) = 1$: peace has higher full-participation power;

(A3) $A'_v(\ell) < A'_p(\ell)$: peaceful power is more responsive to additional participation.

A1 captures the idea that violent tactics can generate some power even with very little mass participation. A2 says that at the opposite end of full mobilization, peaceful tactics are more powerful. A3 ensures that this difference arises systematically as turnout increases.

[2] The tie-breaking assumption is needed only at knife-edge parameter values or histories at which the organizer is exactly indifferent between the two tactics.

Together, the assumptions imply that the two technologies cross exactly once: there exists a unique $\ell^* \in (0,1)$ such that $A_v(\ell) > A_p(\ell) \quad \text{for } \ell < \ell^*$, and vice versa for $\ell > \ell^*$.

Participation. Each potential participant i privately observes a period-t participation cost

$$c_{it} = c_t + \epsilon_{it}$$

where $c_t \sim \text{Unif}[-\epsilon, 1+\epsilon]$ is a common component of participation costs and $\epsilon_{it} \sim \text{Unif}[-\epsilon, \epsilon]$ is idiosyncratic. The common shocks c_t are independent across periods. The idiosyncratic shocks ϵ_{it} are i.i.d. across agents and periods. Citizens do not observe c_t or ϵ_{it} directly. The support of c_t guarantees two-sided dominance and ensures that conditioning on being the marginal participant does not truncate the support of the idiosyncratic shock. In particular, every possible participation cutoff lies at least ϵ away from the boundaries of the common-shock support.

A citizen who participates receives benefit 1 if the protest succeeds and pays their participation cost regardless of the outcome. Thus,

$$u_{it} = \begin{cases} 1 - c_{it}, & \text{if } i \text{ participates and the protest succeeds,} \\ -c_{it}, & \text{if } i \text{ participates and the protest fails,} \\ 0, & \text{if } i \text{ does not participate.} \end{cases}$$

Timing. In each period in which the game remains active:

1. Given the public history and current beliefs F_t, the organizer chooses a tactic $\tau \in \{p, v\}$.
2. Participation costs are realized and privately observed. Citizens simultaneously decide whether to participate.
3. The protest occurs. Turnout ℓ_t and the protest outcome are publicly observed.
4. If $A_\tau(\ell_t) > \theta$, the protest succeeds and the game ends. If the protest fails in period 1, beliefs are updated and the game proceeds to period 2. The game ends after period 2 regardless of the outcome.

The solution concept is Perfect Bayesian Equilibrium.

3.1 Discussion of model assumptions

The model focuses on one channel through which failed protests affect subsequent mobilization: by revealing discouraging information about regime strength, as in AHP. This choice reflects the paper's substantive interest in how organizers anticipate and adapt to endogenous demobilization. In practice, failures may also intensify grievances, encourage organizational learning,

strengthen activists' commitments, or motivate renewed recruitment efforts – potentially increasing future participation.[3] Such mechanisms merit study, but are outside the scope of my research question, which is aimed at understanding how protest movements manage demobilization and decline.

The assumption that peaceful tactics have a higher ceiling of power than violent tactics (A2) reflects the particular forms of contention considered here, in which regime *outsiders* exert pressure on regimes to make changes or to give up power. This excludes phenomena such as coups, which can overthrow even powerful regimes with relatively few participants. Unlike the forms of violent contention considered here, coups typically involve military or political *insiders* who already control parts of the state's coercive apparatus. Their effectiveness therefore depends less on mass participation than on the distribution of power and loyalties within the regime itself.

I also assume that participation costs are independently re-drawn from the same distribution in each period. This abstracts from persistently committed "core activists" and from systematic changes in participation costs over time. For instance, if participation becomes universally more costly after the initial protest, this would compound the demobilizing effects of discouraging information about regime strength, rather than reverse it. Conceptually, a persistent core of committed activists could have an opposite effect to help sustain mobilization despite setbacks. I abstract from these additional dynamics to make the learning–demobilization connection tractable, and to show that the anticipation of declining mobilization can shape tactical choices even without lasting changes in citizens' underlying willingness to participate.

I assume that there are no systematic differences between participation costs associated with each tactic. In practice, tactics may differ in the rewards associated with success, the sanctions associated with failure, or the costs of participation. Violent tactics, for example, may entail greater physical, legal, or psychological risks. I abstract from these differences to isolate the effects of tactical effectiveness: even when violence is no more costly to participate in, and offers greater immediate prospects of success, organizers can optimally begin peacefully. Nevertheless, the model admits a limited reinterpretation in terms of tactic-specific participation costs. Appendix Lemma A1 shows that, for any given belief state, one can hold the tactical production function fixed and introduce tactic-specific cost differences that reproduce the original equilibrium participation cutoffs.[4]

[3] For a formal model in which a committed core left behind by initial failure facilitates subsequent mobilization, see the extension studied in De Mesquita and Shadmehr (2023). The idea that higher regime strength may *increase* the likelihood of successful collective action is also implied by the static model of Shadmehr (2021).

[4] The equivalence is limited: the alternative specification need not preserve equilibrium uniqueness or the learning dynamics studied here.

4 Power and the informational cost of failure

In this section, I abstract away from tactical choice to isolate the feedback loop between protest power, beliefs, and future mobilization. By removing the role of the tactic-choosing organizer, I leave the movement fully exposed to the informational consequences of failure. Because it must use the same protest technology in both periods, it cannot adapt the way it converts participation into power after beliefs and mobilization deteriorate.

The model's core trade-off is that power is a double-edged sword. A more powerful protest is more likely to defeat the regime today, but if it fails, it delivers worse news: only a strong regime could have survived a strong challenge. Formally, a protest succeeds if and only if $A(\ell_t) > \theta$, so after a failure, agents infer that $\theta \geq A(\ell_t)$. The common posterior is thus the prior truncated from below at the realized power of the failed protest.[5]

To show how this change in beliefs weakens the movement's future mobilization capacity, I first characterize its effect on individual participation decisions. Citizens join the protest whenever the expected benefit of success exceeds their private cost. As Lemma 1 shows, the participation decision takes a cutoff form:

Lemma 1 (Cutoff participation)**.** *There exists a threshold $\hat{c}_t$ such that agent i participates in the period-t protest if and only if $c_{it} < \hat{c}_t$, where*

$$\hat{c}_t = \int_0^1 F_t\left(A(\ell)\right) d\ell. \tag{1}$$

Equation (1) gives the first link in the power–learning feedback loop: public beliefs determine the threshold cost above which a citizen is unwilling to join the protest. This threshold is derived by considering the decision of the marginal participant. Intuitively, the expression says that the marginal participant averages over the turnout levels that might be generated, and asks how likely the corresponding protest power is to defeat the regime.[6] Thus, the participation cutoff is simply an average of expected success, given the fixed tactic $A(.)$ and current beliefs F_t.

The next step is to translate the participation cutoff into realized turnout:

[5] This sharp truncation is stronger than the mechanism requires. The key is the "bad news" *direction* of learning. With noisier or more gradual learning, the feedback from failure to beliefs and future mobilization would still operate in the same direction.

[6] This utilizes the standard Laplacian beliefs property of global games (Morris and Shin 2003). Conditional on being marginal, the agent is uncertain only about her rank in the population cost distribution; as the common shock varies uniformly over $(\hat{c}_t - \epsilon, \hat{c}_t + \epsilon)$, her rank – and hence induced turnout – ranges linearly from 0 to 1, yielding a uniform distribution of turnout over $[0, 1]$. The result relies on the uniform distribution of the common shock c_t, but not on uniform idiosyncratic noise – any continuous idiosyncratic distribution yields the same uniform rank belief.

Lemma 2 (Equilibrium turnout)**.** *Given participation cutoff $\hat{c}_t$, realized turnout is*

$$\ell_t = \begin{cases} 1 & \text{if } c_t \leq \hat{c}_t - \epsilon \\ \frac{\hat{c}_t - c_t + \epsilon}{2\epsilon} & \text{if } \hat{c}_t - \epsilon < c_t < \hat{c}_t + \epsilon \\ 0 & \text{if } c_t \geq \hat{c}_t + \epsilon \end{cases} \tag{2}$$

Define

$$\rho := \frac{1}{1 + 2\epsilon} \tag{3}$$

Then $\Pr(\ell_t = 1) = \rho\hat{c}_t$, $\Pr(\ell_t = 0) = \rho(1 - \hat{c}_t)$, *and with remaining probability* $1 - \rho$, *turnout is in* $(0, 1)$ *and is uniformly distributed conditional on being interior. Hence, expected turnout is*

$$\mathbb{E}\left[\ell_t \mid \hat{c}_t\right] = \frac{1 - \rho}{2} + \rho\hat{c}_t \tag{4}$$

Although citizens are forward-looking, their participation decisions are *as if myopic*, as in AHP. Because each citizen is atomistic, her current action cannot affect regime survival or the continuation history, so continuation payoffs cancel from the participation decision.

One feature of the turnout process will be significant for the analysis of tactical choice. As Equation 4 shows, ρ governs how strongly changes in beliefs are amplified into aggregate mobilization. When ρ is high, citizens' participation propensities are tightly clustered around the common shock. The result is that small decreases in $\hat{c}_t$ (driven by changes in contemporaneous beliefs, F_t) can cause many citizens to cross the participation boundary together – significantly affecting turnout. When ρ is low, participation propensities are widely dispersed, so the same belief change produces a more muted aggregate response.

The preceding results characterize behavior at any fixed public belief state, and allow the game to be solved recursively. At every history, public beliefs uniquely determine a participation cutoff (Lemma 1); the cutoff and the common cost shock then uniquely determine turnout and protest power (Lemma 2). Applying these steps yields a unique equilibrium path:

Proposition 1 (Unique cutoff PBE)**.** *There exists a unique Perfect Bayesian Equilibrium in threshold strategies. At every on-path history, agents participate in period t if and only if $c_{it} < \hat{c}_t$ with $\hat{c}_t$ given by* (1)*. Beliefs update by truncation following failure. Given any realization of the common shocks $\{c_t\}_{t=1}^2$, these equilibrium strategies induce a unique path of turnout, protest outcomes, and posterior beliefs.*

The uniqueness result is useful in its own right and does not rely on the two-period horizon. Because atomistic citizens' continuation values cancel from their current participation decisions, the cutoff argument extends to an arbitrary number of periods or an infinite horizon (provided continuation payoffs are well-defined). As AHP show, uniqueness does not follow

simply from repeating a stage game with a unique equilibrium. In their setting, learning from regime survival can eliminate the lower-dominance region after an initial failed attack, restoring equilibrium multiplicity. In my model, the cost-signal structure, together with the support assumption that preserves two-sided dominance after every public history, pins down a unique participation cutoff in every period. The informational consequences of failure can therefore be analyzed without additional equilibrium selection arguments.

Corollary 1 (Failure erodes mobilization)**.** *Following a failed protest,*

$$\hat{c}_{t+1} \leq \hat{c}_t \qquad \textit{and} \qquad \mathbb{E}\left[\ell_{t+1}\right] \leq \mathbb{E}\left[\ell_t\right]$$

where both inequalities are strict whenever failure produces a nontrivial pessimistic update that affects the success probability of the tactic.

The corollary states the dynamic implication of the learning process on mobilization: expected mobilization falls over time. Since a pessimistic shift in current beliefs decreases the participation cutoff, it shrinks the set of citizens whose costs will fall below it. Note, however, that *expected* decline does not preclude *realized* increases in turnout. Because the common shock is re-drawn each period, a particularly low period-2 draw can produce a jump in turnout.

Since the informational cost of failure increases with the power of the failed challenge, should movements sometimes prefer a weaker protest? In the fixed-tactic benchmark, the answer is no. Although more power comes with a higher informational cost upon failure, the two forces do not grow at the same rate. Increasing protest power initially comes with a substantial informational cost: if the protest still fails, the additional power rules out many relatively weak regimes that would otherwise have remained plausible. But this cost tapers off as the protest becomes stronger. Once failure has already selected heavily toward strong regimes, there is progressively less favorable continuation value left for an additional increase in power to destroy, while that additional power continues to expand the set of regimes that can be defeated immediately. In other words, the movement's lifetime probability of success is convex in realized protest power.

Proposition 2 (Higher realized power improves lifetime success)**.** *Fix a single tactic A that is used in both periods. The movement's probability of succeeding by the end of period 2, conditional on realized period-1 power x, is weakly increasing and convex in x.*

The first-order implication of the result is that more power is always better – the informational downside is never large enough to make less power desirable. Convexity offers a second-order implication about the *marginal* value of power: additional power is least valuable when the movement is weak, and becomes increasingly valuable as the protest grows stronger.

Equivalently, holding average first-period power fixed, the movement weakly prefers a lottery over very weak and very strong realizations compared to one concentrated at intermediate levels. The intuition is that intermediate levels of power face the least favorable power-learning trade-off: they may reveal substantial bad news upon failure, without offering the immediate upside of a very strong challenge. This preference for dispersed outcomes becomes particularly relevant in the analysis of tactical choice as $\rho \to 1$ (Section 5.1.2).

Throughout this section, the movement has been fully exposed to the power-learning trade-off. Since the same technology determines both current protest power and the continuation state, there is no way to hedge against or mitigate the damaging effects of failure. The next section introduces tactical choice as precisely such an instrument.

5 Tactical choice

I now study optimal tactical choice and sequencing. Tactics are chosen by the organizer, who is forward-looking, but cannot commit in advance to a tactical sequence: thus, at each point, he evaluates tactics by their effect on the movement's *lifetime* chance of success. For the first period, this means that tactical choice depends upon each tactic's comparative advantage across *future* mobilization states, as well as upon which tactic generates more power today.

Let x denote the realized power of the first-period protest. Schematically, the organizer's $t = 1$ problem is:

$$\max_{\tau_1 \in \{p,v\}} \mathbb{E}_{x|\tau_1} \Bigg[\underbrace{\Pr(\text{success now} \mid x)}_{\text{immediate success}} + \underbrace{\Pr(\text{failure now} \mid x)}_{\text{reach period 2}} \cdot \underbrace{\max_{\tau_2 \in \{p,v\}} \Pr(\text{success later} \mid \text{failure at } x, \tau_2)}_{\text{optimal continuation given mobilization state induced by } x} \Bigg]$$

The expectation is taken over the distribution of first-period power induced by the chosen tactic and equilibrium participation. Following failure, the organizer observes x and chooses the tactic that maximizes the probability of success under the updated beliefs and mobilization conditions.

The organizer's problem resembles a variant of dynamic experimentation. In the typical experimentation setting, a decision-maker chooses among actions or technologies not only for their immediate payoff, but also for what their outcomes reveal, and how that information improves subsequent choices. In my setting, however, experimentation comes with an additional feedback loop: since tactics are turnout-dependent technologies, the information generated by

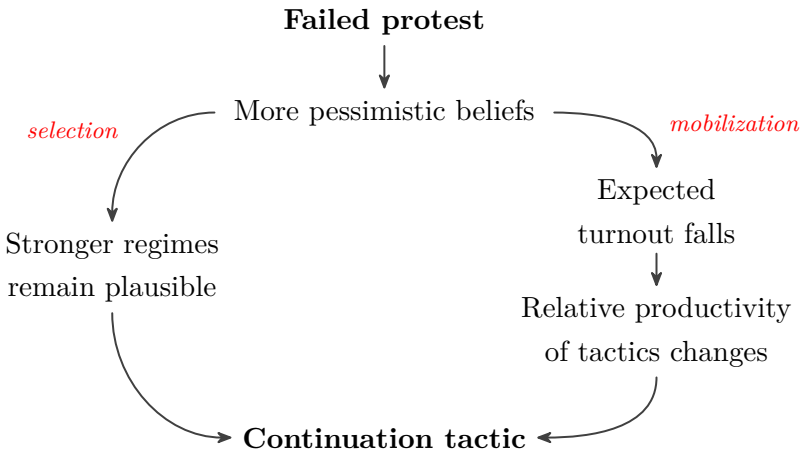


Figure 1. Two channels through which failure shapes tactical choice

a protest changes beliefs, which in turn change the future productivity of the available technologies. Experimentation can therefore degrade the environment in which future exploitation occurs: a failed protest may leave the organizer better informed, but facing lower mobilization and hence a different set of tactically valuable options.

To understand this trade-off, I begin with two observations about the incentives depicted in Figure 1. The *mobilization* incentive implies that a tactic can be worth postponing for future use because it is *not* degraded by learning-from-failure. To see why, suppose an organizer has access to a violent technology which has arbitrarily high baseline power, but no turnout sensitivity at all. The following lemma states that the organizer never uses it first.

Lemma 3 (Flat violence is held in reserve)**.** *If $A_v(\ell) \equiv \bar{v}$ for all ℓ, then a success-maximizing organizer never uses violence in period 1.*

The logic is that of preservation. A completely turnout-insensitive tactic – such as some pre-positioned sabotage that can be triggered without any people on the scene – loses no power from being postponed, as its power is always $\bar{v}$. By contrast, the peaceful tactic is *perishable:* its effectiveness deteriorates with falling turnout. It is thus most valuable at the start of the game, and only weakens (in expectation) after any failure. The organizer therefore uses the perishable tactic first, and preserves the turnout-insensitive tactic for the more pessimistic future which may follow.

As the following result states, violence can attract more participants, generate more expected power and have a higher probability of immediate success, yet still optimally be held in reserve. Its robustness to demobilization creates *option value* that is lost when it is used immediately.

Corollary 2 (Immediate advantage need not imply early use)**.** *Suppose violence is turnout-insensitive, so that $A_v(\ell) \equiv \alpha$, and $\alpha > 1/2$. Then violence attracts greater expected participation, generates greater expected first-period power, and is more likely to succeed immediately than peace. Nevertheless, the organizer strictly prefers to open peacefully.*

My second observation pertains to the selection incentive. Preserving a turnout-insensitive tactic may be valuable from the perspective of the first period, but will it actually be used in the continuation? The answer is not necessarily. Because peace has the higher ceiling of power, sufficiently severe failures can leave behind precisely the strong regimes against which peace has the comparative advantage. A tactic that is valuable to preserve when regime strength is uncertain may cease to be useful once failure has ruled out sufficiently many weaker regimes. The implication is an asymmetry in tactical paths: some movements may optimally remain peaceful after every history, but no movement remains violent after every history (a sufficient condition for peace to be used on the entire path is stated in Appendix Proposition A1.) After a sufficiently powerful failure, only the higher-ceiling peaceful technology remains well suited to the strong regime types left in the posterior.

The remainder of the section uses a linear setting to translate these sequencing forces into concrete predictions about how observable features of movements and their tactical repertoires affect the trajectories that movements actually adopt.

5.1 Linear tactics

In the remainder of the paper, I normalize the peaceful technology to $A_p(\ell) = \ell$, and let

$$A_v(\ell) = \alpha + \beta\ell, \qquad \alpha > 0, \quad \beta \geq 0, \quad \alpha + \beta < 1.$$

The normalization is without loss of generality within the linear setting: the slope of peaceful power simply fixes the units in which protest power and regime strength are measured.

This parameterization makes each tactic's comparative advantage especially transparent. The intercept α captures violence's low-turnout advantage: unlike peace, violence can generate power even with little or no participation. By contrast, the slope difference $1 - \beta$ captures peace's advantage in scaling with turnout. Equivalently, β measures how closely violence keeps pace with peace as additional participants join. Thus, higher α strengthens violence's low-turnout advantage, whereas higher β weakens peace's turnout-based advantage. The restriction $\alpha + \beta < 1$ preserves peace's higher ceiling under full mobilization.

These parameters also lend themselves to empirical interpretation. Violence's baseline capacity α captures the extent to which small groups can generate leverage with low participation. α is high when actors can exert pressure by disrupting strategically or symbolically important

targets, such as infrastructure, centers of state power, or centers of economic activity. Access to capital, rugged terrain, or poorly protected targets can all raise α. Violence's turnout sensitivity β captures the extent to which violent contention itself benefits from numbers. Some forms of violence are inherently manpower-intensive: coordinated assaults, attempts to seize and hold government buildings or territory, sustained barricade defense, and direct-fire engagements become more effective as more participants can be mobilized. Other tactics, such as sabotage, assassination, or some forms of bombing and indirect fire, can be carried out effectively by relatively small groups and therefore depend much less on mass participation.

The linear specification also makes the fundamental objects of the game particularly simple to compute. Under the initial uniform prior,

$$\hat{c}_1^p = \frac{1}{2M}, \qquad \hat{c}_1^v = \frac{\alpha + \beta/2}{M}.$$

Hence, for $\rho > 0$,

$$\mathbb{E}\left[\ell_1^v\right] > \mathbb{E}\left[\ell_1^p\right] \iff \alpha + \frac{\beta}{2} > \frac{1}{2}.$$

Violence can therefore attract *more* participants when its technological advantages make success sufficiently plausible. Peace has no intrinsic participation advantage in the model: relative turnout is itself an equilibrium outcome of tactical effectiveness.[7]

A static benchmark follows immediately. A myopic organizer would simply choose the tactic with the highest immediate probability of success, given by

$$S_1^\tau = \frac{\mathbb{E}\left[A_\tau(\ell_1^\tau)\right]}{M}$$

where ℓ_1^τ denotes equilibrium first-period turnout under tactic τ. Corollary 2 already showed how the dynamic problem can overturn this ranking: even when a tactic attracts more participants, generates more expected power, and is more likely to succeed immediately, it may not be used immediately.

5.1.1 Responses to setbacks

I first characterize tactical choice after a failed protest of size x. A natural conjecture is that more severe setbacks should make violence increasingly attractive: Since these failures have the largest demobilizing effect, they should also make violence – the tactic advantaged at low turnout – most attractive. The model delivers the opposite prediction: when violence is used in continuation, it follows sufficiently *low-power* failures. By contrast, more severe setbacks lead the movement to the peaceful tactic.

[7] Allowing peaceful participation to be intrinsically less costly would mechanically shift participation toward peace and, all else equal, strengthen incentives for peaceful protest. I abstract from this channel to isolate differences in participation generated endogenously by tactical effectiveness and beliefs.

Proposition 3 (Continuation tactics)**.** *There exists a cutoff $c_\rho(\alpha, \beta; M) \in [0, \alpha + \beta)$ such that violence is strictly preferred in period 2 following a failed first-period protest if and only if*

$$x < c_\rho(\alpha, \beta; M).$$

Whenever $c_\rho > 0$, the organizer is indifferent at $x = c_\rho$. The cutoff is weakly increasing in both α and β.

The result is driven by *selection*: larger failures progressively eliminate the weaker regime types against which violence has a comparative advantage. Let ℓ^* denote the turnout level at which the two technologies cross. In the linear specification, $\ell^* = \frac{\alpha}{1-\beta}$, and because $A_p(\ell^*) = A_v(\ell^*) = \ell^*$, this is also the power level associated with the technological crossing. The continuation cutoff satisfies $c_\rho(\alpha, \beta; M) < \ell^*$. Thus, the organizer switches back to peace before realized failure power reaches the power level associated with the technological crossing.

The reason is that tactical choice depends on the entire posterior over regime strength, not just its weakest remaining type. Following a failure of size $x < \ell^*$, violence retains an advantage against the relatively weak regimes with $\theta \in [x, \ell^*)$. But the posterior *also* contains all stronger regimes with $\theta \geq \ell^*$, against which peace requires less turnout to succeed. As x rises toward ℓ^*, the set of regime types for which violence is useful shrinks, while the set for which peace has the advantage remains. Peace therefore becomes optimal strictly before x itself reaches the technological crossing point.

This also bears implications for participation. Once $x \geq \ell^*$, the participation cutoff under peace is weakly higher than under violence (that is, $\hat{c}_2^p(x) \geq \hat{c}_2^v(x)$.) The intuition is simple. At low turnout levels, where violence would normally have a technological advantage, neither tactic now generates enough power to defeat any regime remaining in the posterior. Thus, violence's advantage is irrelevant for citizens' participation decisions. At the higher turnout levels where success remains possible, peace is the stronger technology, and draws more expected participants.

The comparative statics of the cutoff identify which movements are most likely to use violence after a setback. Higher baseline capacity α expands the range of failures after which violence is useful. Higher β does the same for a different reason: as violence becomes more responsive to turnout, peace loses some of its scaling advantage – thus pushing the cutoff point upwards and also expanding the range of failures after which violence is optimal.

Finally, the continuation choice of tactics is not always path dependent. When violent capacity is sufficiently weak – that is, it is not attractive even after the weakest possible failure

– the cutoff collapses to $c_\rho(\alpha, \beta; M) = 0$, so peace is optimal after every possible failure. As violent capacity improves, the cutoff becomes positive and the continuation choice becomes history-dependent.

5.1.2 Choosing an opening tactic

The continuation problem conditions on the power x generated by an initial failed protest. The opening problem is different: before turnout is realized, the organizer's choice of tactic helps determine the *distribution* of first-period power, and hence the set of belief and mobilization states from which the continuation game may begin. The parameter ρ is central to this comparison because it determines how strongly the information revealed by the first protest feeds back into subsequent mobilization. I begin with the two limiting cases of ρ to isolate the mobilization channel clearly.

Resilient mobilization: $\rho \to 0$. When ρ approaches zero, changes in public confidence have vanishingly small effects on realized turnout. Although failure still changes beliefs about regime strength and citizens' willingness to participate, these changes no longer appreciably shift the *distribution* of aggregate mobilization, which approaches $\text{Unif}[0, 1]$ at every belief state. This limiting case therefore serves to shut down the mobilization feedback loop and consider first-period tactical choice in its absence.

	Opening tactic	Following failure
Weak violent capacity	Peace	Peace
Medium violent capacity	Peace	Violence after weak failure; peace after strong failure
Strong violent capacity	Violence	Violence after weak failure; peace after strong failure

Table 1. Equilibrium tactics by violent capacity when $\rho \to 0$

The consequence is that the organizer places greater weight on the advantages of scalable violence under favorable turnout conditions, potentially making a violent opening optimal. The model admits three equilibrium trajectories, summarized in Table 1.[8] To understand the forces that generate these trajectories, I plot the two relevant boundaries in (α, β) space (Figure 2). The orange boundary governs continuation tactical choice, and represents the lowest level of β at which $c_0(\alpha, \beta; M) > 0$, so violence becomes an appealing fallback tactic. The higher green

[8]There is no corresponding region in which the organizer opens with violence but switches to peace after every possible failure. If violence would not remain optimal following even the weakest failed violent opening, then it cannot be strictly optimal to deploy first. Details are given in Appendix B.2.

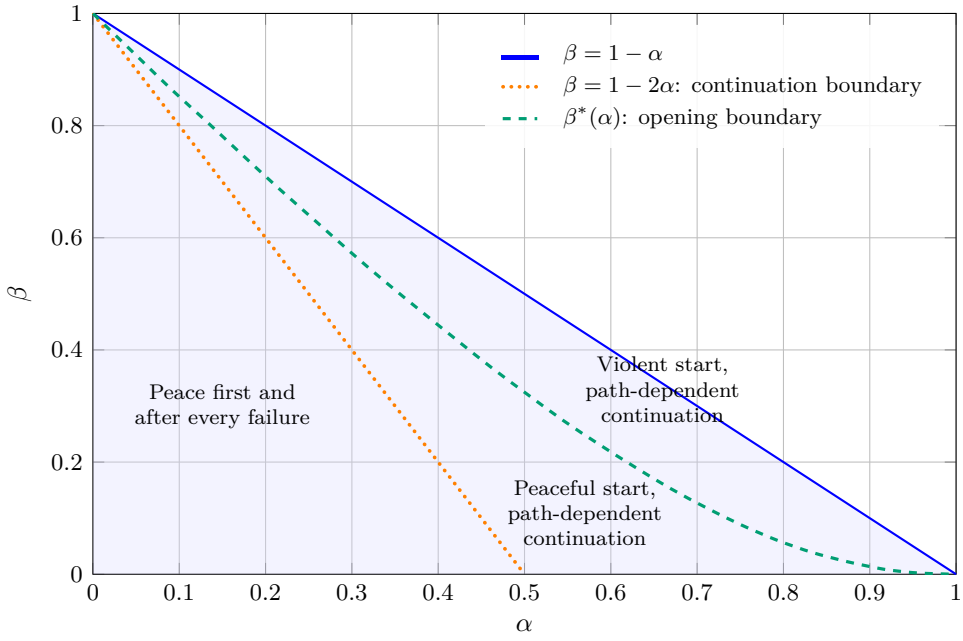


Figure 2. Tactical choice when $\rho \to 0$. The solid blue line marks the boundary of the feasible parameter region, $\beta < 1 - \alpha$, which is shaded blue. The orange dotted line, $\beta = 1 - 2\alpha$, is the continuation boundary: below it, peace is optimal after every possible failure; above it, violence is optimal following sufficiently weak failures. The green dashed line, $\beta^*(\alpha)$, is the opening boundary. It is computed numerically by fixing a grid of values of α and, for each value, solving for the value of β at which the organizer is indifferent between opening peacefully and opening violently, i.e., $W_1^p(\alpha, \beta; M) = W_1^v(\alpha, \beta; M)$.

boundary $\beta^*(\alpha)$ governs the opening decision, and is the $\rho \to 0$ case of the general opening threshold $\beta_\rho^*(\alpha; M)$ characterized in Proposition 4 below.

The figure clarifies how the two dimensions of violent capacity affect the equilibrium trajectory. When violent capacity is sufficiently weak, peace is used throughout. When $\beta = 0$, increasing α only makes violence a useful reserve tactic, but it cannot by itself make violence a useful opening tactic (reproducing Lemma 3). As β increases, violence can become attractive following weak failures, and eventually, it becomes optimal to deploy immediately. Thus, α and β play complementary roles in shifting the movement toward violence, but primarily through different parts of the sequence: α strengthens its fallback role, while β strengthens its appeal under favorable mobilization.

Fragile mobilization: $\rho \to 1$. This case makes the mobilization channel as strong as possible. Since potential participants respond very similarly to even tiny changes in beliefs, realized turnout becomes concentrated near either zero or full mobilization – the interior component of the turnout distribution vanishes. This turnout property means that realized peaceful protest

power approaches an all-or-nothing distribution: the movement generates either essentially no power or the maximum power it can produce.

This produces a threshold-like aggregate technology even though the peaceful production function itself remains linear. Classic threshold accounts of collective action emphasize how clustered participation thresholds can produce abrupt changes in aggregate mobilization (Granovetter 1978; Kuran 1991). A related pattern arises here through the responsiveness of participation to common beliefs: small changes in confidence can move the movement sharply between near-zero and near-full mobilization. Thus, *observed* peaceful power may look highly discontinuous even though the underlying technology is continuous.

This polarization echoes the convexity result in Proposition 2. In the fixed-tactic benchmark, holding average first-period power fixed, greater dispersion improves lifetime success. When $\rho \to 1$, peace generates precisely such polarized outcomes: approximately 0 or 1. It thus offers an unusually attractive opening option. By using peace first, the movement gains all the power upside of a strong challenge (if everyone turns out) and none of the informational downside of a failed weak challenge (if nobody turns out).

Violence, on the other hand, is doubly disadvantaged as an opening tactic, since it compresses realized power toward the interior, approximately α or $\alpha + \beta$. Its baseline capacity $\alpha > 0$ means that even a poorly mobilized violent failure is guaranteed to produce pessimistic information about the regime, while the restriction $\alpha + \beta < 1$ prevents even full violent mobilization from matching peace's upper bound. When changes in beliefs feed strongly into subsequent turnout, this difference in the distribution of first-period power has especially large dynamic consequences.

Note the contrast with the $\rho \to 0$ benchmark. A high β still creates a technological reason to use violence immediately: favorable early mobilization conditions are precisely when violence is most powerful. But when $\rho \to 1$, the preservation motive dominates even at the highest feasible β. Violence can still be optimal following sufficiently weak failures, but it is never used as an opening tactic. The three trajectories from the resilient-mobilization benchmark therefore collapse to two.

Intermediate levels of mobilization. The following proposition generalizes the limiting-case comparison. Above a critical threshold of ρ, violence is never used as an opening tactic, no matter how effective.

Equilibrium trajectory	$\rho \to 0$	$\rho \to 1$	Dependence on ρ
Peace throughout	Possible	Possible	Region shrinks with ρ
Peace first; violence after weak failures, peace after strong failures	Possible	Possible	Region expands in between extreme values of ρ
Violence first; violence after weak failures, peace after strong failures	Possible	Absent	Region disappears above a critical ρ

Table 2. Equilibrium tactical trajectories and mobilization responsiveness.

Proposition 4 (Mobilization responsiveness and violent openings)**.** *For every (α, M), there exists a unique threshold $\bar{\rho}(\alpha, M) \in (0, 1)$ such that:*

(i) If $\rho < \bar{\rho}(\alpha, M)$, sufficiently scalable violence is used first. In particular, there exists a unique threshold $\beta_\rho^*(\alpha; M) < 1 - \alpha$ such that the organizer opens violently if and only if

$$\beta > \beta_\rho^*(\alpha; M).$$

(ii) If $\rho \geq \bar{\rho}(\alpha, M)$, the organizer opens peacefully for every admissible violent technology.

Thus, holding the tactical repertoire fixed, changes in the responsiveness of mass participation can generate discrete changes in the trajectories available to a movement, and in particular in whether violence can appear at the outset.

Table 2 summarizes how mobilization responsiveness reshapes equilibrium tactical trajectories. As ρ increases, the region in which movements remain exclusively peaceful shrinks, because tactics that retain leverage under low turnout become more attractive as fallbacks. At the same time, the region in which movements open violently eventually disappears: even highly effective violent tactics become more valuable to preserve for later use. Together, these forces expand the range of circumstances in which movements begin peacefully but turn to violence following sufficiently weak failures.

5.1.3 Structural change and tactical trajectories

The preceding results offer a theory of how broader structural changes reshape the trajectories of contention. As scholars of comparative politics have long observed, urbanization, changes in infrastructure and communications, transformations of public space, and shifts in the social composition of movements all shape movements' repertoires of contention – and in particular, how effectively different tactics convert turnout into political leverage. The model implies that

such changes can affect not only which tactics are viable, but *when* they are used.

Urbanization provides the clearest example. The movement of revolutionary contention toward cities concentrated populations, economic activity, and centers of political authority in the same space, while also bringing challengers into closer contact with the coercive apparatus of the state (Beissinger 2022; Wallace 2014). One plausible consequence is to *rotate* the violent technology (Kalyvas and Balcells 2010): urbanization reduces the leverage that a small armed group can exercise independently of mass support (Fearon and Laitin 2003), while increasing the extent to which coercive tactics benefit from large numbers (Chenoweth and Stephan 2011; Staniland 2010). In the language of the linear setting, α falls while β rises.

The effect on trajectories is a rise in early-stage violence. A decline in α erodes violence's comparative advantage after demobilization: weakly attended movements can no longer generate as much leverage simply by switching tactics. An increase in β makes violent confrontation increasingly effective when turnout is already high. Urbanization can therefore shift violence *forward* in the tactical sequence even while reducing its value as a fallback.

The reconstruction of Paris under Haussmann is a localized example of the same logic. The narrow streets of the old city made barricades comparatively effective instruments for small groups of insurgents (Traugott 2010). After the Haussmannian renovation, broad boulevards and large, open spaces facilitated troop movement and rendered barricade insurrection more difficult and less effective (Hobsbawm 2005; Traugott 2010). From the perspective of the model, this meant that successful street confrontation could no longer be easily sustained by a small group of insurgents defending barricades, and required assembling and sustaining larger numbers. Thus, to the extent that the transformation both lowers α and raises the relative importance of β, it should reduce the incidence of violence as a post-demobilization fallback and concentrate whatever violent contention remains in periods when mobilization is already broad.

The development of critical infrastructure can move the tactical environment in the opposite direction. Rail lines, energy systems, communications networks, ports, and other concentrated nodes can create opportunities for a relatively small number of actors to impose costs far beyond their numbers (Mitchell 2011; Silver 2003; Wright 2000). Where such systems are vulnerable to disruption (Tichý and Eichler 2018), technological development can raise α without comparably increasing β. The model predicts that such changes should move violence *backward* in the sequence. Higher baseline capacity expands the set of post-failure states in which violence is useful, making peaceful-to-violent transitions more likely after weak setbacks. But precisely because this capacity survives demobilization, it need not be spent while participation is high. The proliferation of vulnerable strategic targets can therefore increase a movement's capacity

for violence while making that capacity disproportionately visible *after* mass mobilization falters. Decentralizing or protecting those same nodes should have the opposite effect: it removes violence's low-turnout insurance value and shrinks the region of peaceful-to-violent switching.

Structural change can also reshape the movement's mobilization base itself: how much do supporters differ in their response to changing prospects of success? Durable parties, unions, neighborhood organizations, and activist networks can produce a layered constituency, where not all citizens respond identically to disappointments. Such heterogeneity dampens the aggregate response to any common deterioration in confidence, corresponding to a lower ρ. The opposite environment can produce much sharper swings in mobilization. This possibility is especially familiar from work on protest under authoritarian rule, where repression and preference falsification can leave citizens uncertain about both the extent of latent opposition and the regime's vulnerability (Kuran 1991; Lohmann 1994). When participation depends heavily on these common assessments, a single political signal can move large numbers of otherwise similar citizens across the participation margin at once. What may appear empirically as a volatile "mass psychology" can therefore arise from highly synchronized responses to common information, rather than from unusually fickle individual preferences. Hence, "high ρ" movements can surge rapidly when success appears plausible, but can also collapse rapidly following discouraging failures.

6 Repression

I have so far treated the movement's tactical environment as fixed. In practice, states actively intervene in that environment. In this section, I study how repression interacts with two forces at the center of the baseline model: tactical substitution and learning from failure. While characterizing the regime's fully unrestricted dynamic repression policy is outside the scope of this study, I analyze two benchmark policy problems, each designed to isolate one of these channels. The first studies how the movement's ability to substitute across tactics constrains the effects of responsive repression – that is, measures taken after an initial protest has already occurred. The second holds responsive repression fixed in order to isolate how preemptive repression – enacted before any protest has happened – changes the impact of failures on future mobilization.

Formally, I introduce a regime actor whose objective opposes that of the organizer: the regime receives payoff 1 if it survives the movement and 0 if the movement succeeds. In each period, the regime can pay some cost to weaken tactic τ by a factor $\gamma_t^\tau \in [0, 1]$, so that realized protest power is $x_t = \gamma_t^\tau A_\tau(\ell_t)$, (so lower γ denotes more intense repression.) I make several assumptions: First, repression is publicly observed before participation decisions, so citizens account for it both when deciding whether to participate and when interpreting a subsequent

failure. Second, I focus on *tactic-specific* interventions which affect one tactic, and leave the alternative technology unchanged. Finally, I assume that the regime shares the movement's uncertainty about its underlying strength.[9]

I begin by analyzing the limits to responsive repression enacted in the terminal period. At this point, the regime only targets the tactic which the protest will use. As repression progressively weakens this tactic, however, the alternative eventually becomes more attractive. Once the organizer switches, further repression of the abandoned tactic has no additional effect on tactical choice or the movement's probability of success – effectively placing a cap on the value of repression.

Proposition 5 (Repression and tactical substitution)**.** *Suppose that, following some first-period failure, the organizer strictly prefers one tactic while the alternative tactic remains viable. If the regime represses only the preferred tactic, there is a unique repression threshold at which the organizer switches to the alternative. Once that threshold is reached, further repression of the original tactic cannot further reduce the movement's probability of success.*

Notably, this demonstrates one reason why tactical adjustment patterns may differ from those described in the previous section. Absent repression, weak failures lead toward violence while stronger failures lead toward peace. Repressing violence following a weak failure can reverse this logic by inducing a switch toward peace. Repressing peace following a stronger failure can analogously redirect the movement toward violence whenever violence remains a viable fallback. Thus repression can generate *opposite patterns* of tactical adjustment from pure learning.

I now turn to *preemptive* repression enacted before the first protest. Such intervention weakens the power of the movement's immediate challenge, but *also* changes what that failure reveals about regime strength. A weaker failure can therefore leave the movement more optimistic, and better able to mobilize, in the period that follows. To isolate this informational consequence, I hold period-2 repression fixed at zero. The previous subsection already characterizes how the regime can respond once a failure has occurred; this benchmark isolates how preemptive repression changes the state from which that continuation begins. Because weaker failures leave movements more optimistic, repression consequently returns part of its immediate benefit to the movement through more favorable continuation beliefs:

Proposition 6 (The belief refund)**.** *Suppose period-2 technologies are unrepressed. Lowering realized first-period protest power weakly improves the movement's continuation prospects. Moreover, for sufficiently weak first-period challenges ($x \leq \min\{\alpha, c_\rho(\alpha, \beta; M)\}$), lowering*

[9]Unobserved repression or private regime information would introduce additional inference and signaling dynamics that I abstract from here.

first-period power leaves the movement's eventual probability of success unchanged when $\rho\beta = 0$*, but strictly* increases *eventual success when* $\rho > 0$ *and* $\beta > 0$*.*

The proposition is conditional on realized first-period power. It characterizes the informational consequences of making a protest weaker, rather than the organizer's ex ante response to a particular repression policy, which also changes the distribution of first-period outcomes.

The belief refund can do more than erase the regime's immediate gain from repression: when mobilization responds to beliefs, it can make repression *counterproductive*. The contrast is clearest in the two limiting cases. When $\rho = 0$, turnout is insensitive to beliefs, and repression can receive a *full refund* in the region of sufficiently weak first-period challenges. In this region, failure induces a switch to violence, whose baseline capacity is sufficient to defeat the marginal regime types spared by repression in period 1. When $\rho > 0$, however, repression can protect and even bolster mobilization. By weakening the first-period challenge, the regime makes a subsequent failure less discouraging. Because turnout responds to beliefs, the resulting optimism translates into higher second-period participation. Crucially, that additional participation can make the movement's fallback tactic more powerful. When $\beta > 0$, violence becomes more effective as more citizens participate. If turnout is high, the violent tactic can defeat the marginal regime types that repression spared in period 1, and even threaten still stronger regimes. The refund is then *more than complete*: reducing the movement's immediate power can increase its eventual probability of success.

This is the regime-side counterpart of the preservation motive in the previous section. There, a forward-looking organizer values robust violence because failure destroys the mobilization conditions under which peaceful protest is effective. Preemptive repression partly prevents that destruction: by weakening a failure, the regime actually saves the optimism and mobilization that the movement would otherwise have lost.

The same informational logic has a direct implication for the persistence of violence. Consider a regime that represses an anticipated violent opening. As long as the organizer continues to open violently, stronger counterinsurgency makes the initial protest both more likely to fail, and to fail *weakly*. Weak failures generate precisely the states in which violence remains optimal in the continuation.

Proposition 7 (Weakening versus displacing violence)**.** *Suppose repression affects only the period-1 violent technology and there is no period-2 repression. Among repression policies under which the organizer continues to open violently, stronger repression weakly increases the expected number of periods of violent protest. By contrast, any policy strong enough to induce a peaceful opening produces strictly less expected violence than any policy that preserves a violent opening.*

Repressing violence can therefore have different effects depending on whether it actually prompts the movement to change course. If repression is not too strong and the movement continues to begin violently, stronger repression shifts failures toward precisely the continuation states in which violence recurs. Once repression becomes strong enough to displace the violent opening, the certain first-period episode of violence disappears.

7 Discussion

This paper develops a model of dynamic tactical choice in repeated protests. Its broader contribution is to identify a form of political experimentation in which experimentation is informative about the environment but also changes the productivity of the actions available to the decision-maker. Here, protest outcomes reveal information about regime strength, but also feed back into subsequent mobilization, which in turn determines the effectiveness of the tactics available to the movement. Experimentation can therefore erode the value of some technologies while preserving others. This creates incentives to hold robust tactics in reserve, to switch tactics following some failures but not others, and even to forgo the tactic with the greatest immediate prospect of success. The relevant dynamic tradeoff is thus not simply between current success and future information, but between current success and the political environment in which future contention will take place.

The model also suggests a different interpretation of the empirical success advantage of nonviolent resistance (Chenoweth and Stephan 2011). This advantage is commonly attributed in part to the greater participation that nonviolent tactics can sustain. My model identifies a less obvious implication of this dependence on participation: organizers may prefer to begin peacefully precisely *because* mass-based nonviolent tactics are vulnerable to demobilization caused by past failures – even if they are not inherently more effective. This bears implications for selection *into* observed tactics: some movements succeed peacefully before violence is ever deployed, while violence held in reserve is more likely to be observed after an earlier challenge has failed and mobilization has deteriorated. The apparent advantage of peaceful resistance may therefore reflect not only its effectiveness as a tactic, but also the strategic value of deploying participation-intensive tactics early, while preserving more robust alternatives for later.

Repression introduces a second strategic actor into this problem. State intervention aimed at reducing the immediate effectiveness of a protest also changes the information generated by failure, and alters the movement's subsequent tactical choices. This helps explain why the regime's ability to weaken a particular protest need not translate into greater long-run security, and why flexibility in responding to future contention can on its own change the movement's incentives in the present.

A fuller analysis of the regime's intertemporal problem presents one avenue for future work. In particular, greater policy flexibility need not benefit the regime, because movements choose tactics in anticipation of future state responses. If an organizer expects violence to be heavily repressed once it becomes an attractive fallback after failure, the value of preserving violence for that contingency falls, potentially inducing the movement to deploy it earlier. A regime that could credibly commit to future forbearance might therefore induce a peaceful opening by relinquishing an intervention it would prefer to use ex post. This raises a broader commitment problem: restrictions on the regime's future actions may improve its position today by changing the movement's incentives over how to sequence tactics.

The same informational logic also suggests a role for the public transmission of protest outcomes. If failed protests demobilize partly through what they reveal about regime strength, then obscuring the scale of a strong but unsuccessful challenge can blunt the discouraging information generated by its failure. Conversely, a regime may sometimes benefit from allowing opposition setbacks to become widely observed, even when greater transparency also reveals the strength of the challenge itself. A natural extension would therefore allow the regime to shape the precision or dissemination of information about protest outcomes, connecting the model to work on information control and the informational foundations of repression.

The institutional organization of tactical choice provides another extension. The unitary organizer studied here offers a useful benchmark, but contested leadership and collective decision-making may substantially alter the incentives to experiment with, preserve, or switch between tactics. With multiple leaders, tactical sequences may depend on bargaining power, veto rights, opportunities for unilateral action, and the possibility of organizational fragmentation. Under more decentralized forms of decision-making, tactical choice may instead depend on changes in the composition of the movement itself, as different constituencies enter, exit, or gain influence following success and failure. Extending the analysis to these alternative decision structures would help clarify how organizational institutions mediate the dynamic incentives identified here.

More generally, the model points toward a class of experimentation problems in which actions are technologies whose productivity depends on an endogenous state that past outcomes themselves reshape. Protest tactics provide one instance, but similar forces may arise whenever political actors choose among instruments that differ in their dependence on participation, organizational capacity, public support, or other resources that political experience can subsequently erode or replenish. A broader theory of such problems could characterize these incentives in a more general experimental environment, extending some of the insights

developed in the more specific case studied here. Protest tactics can thus be seen as a special case of a general dynamic problem in which actors choose amongst technologies whose future value depends on the conditions that today's actions help create.

AI Use Declaration

During the preparation of this manuscript (2024–2026), I used OpenAI's ChatGPT (including GPT-5.6 and GPT-6) to assist with refining the exposition and interpretation of the theoretical model, checking mathematical derivations and proof arguments, and preparing LaTeX code for tables and figures. AI-generated suggestions were reviewed, revised, and selectively incorporated into the manuscript by hand. I retain full responsibility for the theoretical results, arguments, citations, and final manuscript.

References

Angeletos, George-Marios, Christian Hellwig and Alessandro Pavan. 2007. "Dynamic global games of regime change: Learning, multiplicity, and the timing of attacks." *Econometrica* 75(3):711–756.

Beissinger, Mark R. 2022. *The Revolutionary City: Urbanization and the Global Transformation of Rebellion*. Princeton University Press.

Blair, Christopher W. 2024. "The fortification dilemma: Border control and rebel violence." *American Journal of Political Science* 68(4):1366–1385.

Boix, Carles and Milan W. Svolik. 2013. "The Foundations of Limited Authoritarian Government: Institutions, Commitment, and Power-Sharing in Dictatorships." *The Journal of Politics* 75(2):300–316.

Boyer, Pierre C, Germain Gauthier, Yves Le Yaouanq, Vincent Rollet and Benoît Schmutz-Bloch. 2026. The lifecycle of protests in the digital age. Technical report Working Paper.

Bueno de Mesquita, Ethan. 2013. "Rebel tactics." *Journal of Political Economy* 121(2):323–357.

Casper, Brett Allen and Scott A. Tyson. 2014. "Popular Protest and Elite Coordination in a Coup d'état." *The Journal of Politics* 76(2):548–564.

Chenoweth, Erica and Maria J Stephan. 2011. *Why civil resistance works: The strategic logic of nonviolent conflict*. Columbia University Press.

Dasgupta, Amil. 2007. "Coordination and delay in global games." *Journal of Economic Theory* 134(1):195–225.

Dasgupta, Amil, Jakub Steiner and Colin Stewart. 2012. "Dynamic coordination with individual learning." *Games and Economic Behavior* 74(1):83–101.

De Mesquita, Ethan Bueno and Mehdi Shadmehr. 2023. "Rebel motivations and repression." *American Political Science Review* 117(2):734–750.

della Porta, Donatella. 1995. *Social Movements, Political Violence, and the State: A Comparative Analysis of Italy and Germany.* Cambridge: Cambridge University Press.

Dube, Oeindrila and Juan F Vargas. 2013. "Commodity price shocks and civil conflict: Evidence from Colombia." *Review of Economic studies* 80(4):1384–1421.

Fearon, James D and David D Laitin. 2003. "Ethnicity, insurgency, and civil war." *American political science review* 97(1):75–90.

Gieczewski, Germán. 2021. "Policy persistence and drift in organizations." *Econometrica* 89(1):251–279.

Gieczewski, Germán and Korhan Koçak. 2024. "Collective Procrastination and Protest Cycles." *American Journal of Political Science (Forthcoming)* .

Gieczewski, Germán and Svetlana Kosterina. 2024. "Experimentation in Endogenous Organizations." *Review of Economic Studies* 91(3):1711–1745.

Granovetter, Mark. 1978. "Threshold models of collective behavior." *American journal of sociology* 83(6):1420–1443.

Hobsbawm, Eric J. 2005. "Cities and insurrections." *Global Urban Development* 1(1):1–8.

Inclán, María. 2018. *The Zapatista Movement and Mexico's Democratic Transition: Mobilization, Success, and Survival.* Oxford University Press.

Kalyvas, Stathis N and Laia Balcells. 2010. "International system and technologies of rebellion: How the end of the Cold War shaped internal conflict." *American Political Science Review* 104(3):415–429.

Kuran, Timur. 1991. "Now out of never: The element of surprise in the East European revolution of 1989." *World politics* 44(1):7–48.

Little, Andrew T. 2017. "Coordination, learning, and coups." *Journal of Conflict Resolution* 61(1):204–234.

Little, Andrew T., Joshua A. Tucker and Tom LaGatta. 2015. "Elections, Protest, and Alternation of Power." *The Journal of Politics* 77(4):1142–1156.

Lohmann, Susanne. 1994. "The dynamics of informational cascades: The Monday demonstrations in Leipzig, East Germany, 1989–91." *World Politics* 47(1):42–101.

Mandela, Nelson. 1964. "Statement from the Dock at the Opening of the Defence Case in the Rivonia Trial." Pretoria Supreme Court, 20 April 1964. Text available from the Nelson Mandela Foundation.

Manekin, Devorah and Tamar Mitts. 2022. "Effective for whom? Ethnic identity and nonviolent resistance." *American Political Science Review* 116(1):161–180.

Mathevet, Laurent and Jakub Steiner. 2013. "Tractable dynamic global games and applications." *Journal of Economic Theory* 148(6):2583–2619.

McAdam, Doug. 1983. "Tactical innovation and the pace of insurgency." *American sociological review* pp. 735–754.

McCauley, Clark and Sophia Moskalenko. 2008. "Mechanisms of Political Radicalization: Pathways toward Terrorism." *Terrorism and Political Violence* 20(3):415–433.

Mitchell, Timothy. 2011. "Carbon Democracy: Political Power in the Age of Oil.".

Morris, Stephen and Hyun Song Shin. 1998. "Unique equilibrium in a model of self-fulfilling currency attacks." *American Economic Review* pp. 587–597.

Morris, Stephen and Hyun Song Shin. 2003. Global games: Theory and applications. In *Advances in Economics and Econometrics: Theory and Applications, Eighth World Congress, Volume 1*. Cambridge University Press pp. 56–114.

Morris, Stephen and Mehdi Shadmehr. 2024. "Repression and Repertoires." *American Economic Review: Insights (Forthcoming)* .

Qiu, Xiaoyan. 2022. "Rebel strategies and the prospects for peace." *American Journal of Political Science* 66(1):140–155.

Ritter, Emily Hencken and Courtenay R. Conrad. 2016. "Preventing and Responding to Dissent: The Observational Challenges of Explaining Strategic Repression." *American Political Science Review* 110(1):85–99.

Shadmehr, Mehdi. 2019. "Multiplicity and uniqueness in regime change games." *The Journal of Politics* 81(1):303–308.

Shadmehr, Mehdi. 2021. "Protest puzzles: Tullock's paradox, hong kong experiment, and the strength of weak states." *Quarterly Journal of Political Science* 16(3):245–264.

Shadmehr, Mehdi and Dan Bernhardt. 2017. "When Can Citizen Communication Hinder Successful Revolution?" *Quarterly Journal of Political Science* 12(3):301–323.

Silver, Beverly. 2003. *Forces of labor: Workers' Movements and Globalization Since 1870*. Cambridge University Press.

Simpson, Brent, Robb Willer and Matthew Feinberg. 2018. "Does violent protest backfire? Testing a theory of public reactions to activist violence." *Socius* 4:2378023118803189.

Sonin, Konstantin and Austin L Wright. 2024. "Rebel capacity, intelligence gathering, and combat tactics." *American Journal of Political Science* 68(2):459–477.

Staniland, Paul. 2010. "Cities on fire: Social mobilization, state policy, and urban insurgency." *Comparative Political Studies* 43(12):1623–1649.

Steinert-Threlkeld, Zachary, Alexander Chan and Jungseock Joo. 2022. "How state and protester violence affect protest dynamics." *The Journal of Politics* 84(2):798–813.

Tichý, Lukáš and Jan Eichler. 2018. "Terrorist attacks on the energy sector: The case of Al Qaeda and the Islamic state." *Studies in Conflict & Terrorism* 41(6):450–473.

Tilly, Charles. 1978. *From Mobilization to Revolution*. Addison-Wesley.

Traugott, Mark. 2010. *The insurgent barricade*. Univ of California Press.

Wallace, Jeremy L. 2014. *Cities and stability: Urbanization, redistribution, and regime survival in China*. Oxford University Press.

Wasow, Omar. 2020. "Agenda seeding: How 1960s black protests moved elites, public opinion and voting." *American Political Science Review* 114(3):638–659.

Wright, Erik Olin. 2000. "Working-class power, capitalist-class interests, and class compromise." *American journal of sociology* 105(4):957–1002.

Appendix

Table of Contents

A Proofs for participation and benchmark results **A1**

A.1 Participation and turnout A1

A.2 Participation-equivalent cost representation A4

A.3 Realized power and lifetime success A5

A.4 Benchmark results with tactical choice A6

B Linear setting **A8**

B.1 Continuation choice (Proposition 3) A8

B.2 The opening comparison A11

B.3 Proof of Proposition 4 . . . . . A14

B.4 Algebraic verifications A16

C Proofs for repression extension **A20**

C.1 Responsive repression A20

C.2 Preemptive repression and the belief refund A22

C.3 Persistence of violence A24

A Proofs for participation and benchmark results

A.1 Participation and turnout

Proof of Lemma 1. Fix beliefs F_t and a tactic with production function A. Because each citizen is atomistic, her current participation decision has no effect on whether the regime survives, the posterior inherited by the next period, or the continuation game. Continuation payoffs therefore cancel from the comparison between participating and abstaining, so participation is as-if myopic.

Suppose other citizens use cutoff $\hat{c}_t$. A higher realization of c_{it} shifts citizen i's posterior over the common cost c_t upward, which weakly lowers expected turnout and hence the expected benefit from participating. Since the cost of participation also rises one-for-one in c_{it}, the payoff difference between participating and abstaining is strictly decreasing in c_{it}. It is therefore enough to characterize the marginal citizen who is indifferent at $c_{it} = \hat{c}_t$.

Conditional on $c_{it} = \hat{c}_t$, $c_t = \hat{c}_t - \epsilon_{it}$. Since $\hat{c}_t \in [0, 1]$ and $c_t \sim \text{Unif}[-\varepsilon, 1 + \varepsilon]$, the interval $[\hat{c}_t - \varepsilon, \hat{c}_t + \varepsilon]$ lies inside the support of the common shock. Conditioning on marginality therefore does not truncate the support of ϵ_{it}. Moreover, because both c_t and ϵ_{it} have constant densities on their supports, the marginal citizen's posterior over c_t is uniform on $[\hat{c}_t - \varepsilon, \hat{c}_t + \varepsilon]$.

For any c_t in this interval, aggregate turnout under cutoff $\hat{c}_t$ is

$$\ell(c_t; \hat{c}_t) = \Pr(c_t + \epsilon_{jt} < \hat{c}_t) = \frac{\hat{c}_t - c_t + \varepsilon}{2\varepsilon}.$$

Thus, as the marginal citizen's posterior over c_t ranges uniformly over $[\hat{c}_t - \varepsilon, \hat{c}_t + \varepsilon]$, induced turnout ranges uniformly over $[0, 1]$. Her expected benefit from participating is therefore

$$\int_0^1 F_t(A(\ell))\, d\ell.$$

Indifference of the marginal citizen gives

$$\hat{c}_t = \int_0^1 F_t(A(\ell))\, d\ell,$$

which is equation (1). Since the right-hand side lies in $[0, 1]$, it defines a valid cutoff; by the monotonicity above, citizens with $c_{it} < \hat{c}_t$ participate and those with $c_{it} > \hat{c}_t$ abstain. □

Proof of Lemma 2. Fix a participation cutoff $\hat{c}_t \in [0,1]$. Actual turnout is

$$\ell_t = \begin{cases} 1, & c_t \le \hat{c}_t - \varepsilon, \\ \dfrac{\hat{c}_t - c_t + \varepsilon}{2\varepsilon}, & \hat{c}_t - \varepsilon < c_t < \hat{c}_t + \varepsilon, \\ 0, & c_t \ge \hat{c}_t + \varepsilon. \end{cases}$$

The support of c_t has length $1 + 2\varepsilon$. The set of common shocks generating full turnout has length $\hat{c}_t$, so

$$\Pr(\ell_t = 1) = \frac{\hat{c}_t}{1 + 2\varepsilon} = \rho \hat{c}_t.$$

The set generating zero turnout has length $1 - \hat{c}_t$, so

$$\Pr(\ell_t = 0) = \rho(1 - \hat{c}_t).$$

The remaining interval has length 2ε and hence probability $1 - \rho$. Conditional on this interior region, the mapping from c_t to ℓ_t is linear, so realized turnout is uniformly distributed on $(0,1)$. Consequently,

$$\mathbb{E}[\ell_t \mid \hat{c}_t] = \rho \hat{c}_t + (1 - \rho)\frac{1}{2} = \frac{1 - \rho}{2} + \rho \hat{c}_t.$$

□

Lemma 2 implies that for a fixed tactic A, its probability of success under beliefs F_t is defined as follows:

Definition A1 (Equilibrium success probability)**.** *For beliefs F_t and tactic A, denote by $S(A, F_t)$ the actual probability of success. This is given by:*

$$S(A, F_t) = \rho(1 - \hat{c}_t) F_t(A(0)) + (1 - \rho) \int_0^1 F_t(A(\ell))\, d\ell + \rho \hat{c}_t F_t(A(1)).$$

Since

$$\hat{c}_t = \int_0^1 F_t(A(\ell))\, d\ell,$$

this can equivalently be written as

$$S(A, F_t) = \rho(1 - \hat{c}_t) F_t(A(0)) + (1 - \rho)\hat{c}_t + \rho \hat{c}_t F_t(A(1)).$$

Proof of Proposition 1. At every history, beliefs F_t and the tactic in use uniquely determine citizens' participation cutoff through equation (1). Given this cutoff and a realization of the common shock c_t, Lemma 2 uniquely determines realized turnout and therefore realized protest power. Following a failed protest, Bayes' rule uniquely determines the truncated posterior.

Backward induction therefore uniquely determines citizens' threshold strategies and the organizer's optimal tactic at every history, with indifference resolved by the tie-breaking convention in favor of peace. Hence, for every realization of the underlying state and common shocks, equilibrium strategies induce a unique path of turnout, protest outcomes, and posterior beliefs. □

Proof of Corollary 1. Suppose a period-t protest fails at realized power x. The posterior F_{t+1} is obtained by truncating F_t from below at x. Hence

$$F_{t+1}(y) \leq F_t(y) \qquad \text{for every } y,$$

with strict inequality on a nonempty interval whenever the failure produces a nontrivial update. Equation (1) then implies

$$\hat{c}_{t+1} = \int_0^1 F_{t+1}(A(\ell))\, d\ell \leq \int_0^1 F_t(A(\ell))\, d\ell = \hat{c}_t.$$

The inequality is strict whenever the posterior shift affects a positive-measure set of power realizations generated by the tactic.

By Lemma 2, expected turnout is

$$\mathbb{E}[\ell_t \mid \hat{c}_t] = \frac{1-\rho}{2} + \rho \hat{c}_t.$$

which is strictly increasing in $\hat{c}_t$. The result therefore follows immediately from the statement that $\hat{c}_{t+1} \leq \hat{c}_t$. □

A.2 Participation-equivalent cost representation

Lemma A1 (Participation-cutoff representation)**.** *Fix a belief state F_t, and let $\hat{c}_t^p$ and $\hat{c}_t^v$ denote the equilibrium participation cutoffs under the two tactical technologies.*

Define

$$k_p(F_t) = 0, \qquad k_v(F_t) = \hat{c}_t^p - \hat{c}_t^v.$$

Consider an alternative specification in which both tactics use technology A_p, but participation under tactic τ entails cost $c_{it} + k_\tau(F_t)$.

This specification admits a threshold equilibrium that reproduces the original participation cutoffs and turnout distributions, conditional on F_t and the chosen tactic.

Proof. In the original model, Lemma 1 establishes that

$$\hat{c}_t^\tau = \int_0^1 F_t(A_\tau(\ell))\, d\ell.$$

In the alternative specification, let $\tilde{c}_t^\tau$ denote the equilibrium cutoff under tactic τ. Since both tactics now use A_p, the marginal citizen's indifference condition is

$$\tilde{c}_t^\tau + k_\tau(F_t) = \int_0^1 F_t(A_p(\ell))\, d\ell = \hat{c}_t^p.$$

Therefore,

$$\tilde{c}_t^p = \hat{c}_t^p,$$

and

$$\tilde{c}_t^v = \hat{c}_t^p - k_v(F_t) = \hat{c}_t^p - (\hat{c}_t^p - \hat{c}_t^v) = \hat{c}_t^v.$$

Thus the equilibrium participation cutoffs are identical under the two specifications. □

The implied cost differential has a simple interpretation: if violence offers a lower expected probability of success than peace, then $\hat{c}_t^v < \hat{c}_t^p$, so $k_v(F_t) > 0$: from the perspective of individual participation decisions, violence behaves as though it were more costly. Conversely, if violence offers a higher expected probability of success, then $k_v(F_t) < 0$, and violence behaves as though it were less costly. Thus, differences in tactical effectiveness can be interpreted as implicit participation-cost differences, although the sign and magnitude of these differences depend on prevailing beliefs.

As a caveat, this result establishes the existence of the participation-equivalent threshold equilibrium. However, it need not preserve two-sided limit dominance or equilibrium uniqueness, and may not generate the same protest outcomes or subsequent beliefs: tactic-specific technologies directly affect realized protest power, whereas participation costs operate through citizens' decisions to participate. The equivalence therefore does not extend to the organizer's dynamic optimization problem.

A.3 Realized power and lifetime success

Proof of Proposition 2. Fix a single increasing tactic A that is used in both periods, and let $\bar{A} := A(1)$. Let $V(x)$ denote the movement's probability of succeeding by the end of period 2 conditional on realized period-1 power x.

Recall that after a failure at power x, the posterior is Unif$[x, M]$. Define

$$h(x) := \int_0^1 (A(\ell) - x)_+ \, d\ell, \qquad G(x) := \int_0^1 \mathbf{1}\{A(\ell) > x\} \, d\ell.$$

Intuitively, $h(x)$ measures the tactic's average power above x, while $G(x)$ is the share of turnout realizations for which the tactic generates power above x. Note also that function h is absolutely continuous, with $h'(x) = -G(x)$ almost everywhere.

Recall that the period-2 participation cutoff is given by

$$\hat{c}_2(x) = \frac{h(x)}{M - x}.$$

Full turnout succeeds with probability

$$r(x) := \frac{\bar{A} - x}{M - x}.$$

Using Lemma 2, the actual period-2 success probability is therefore

$$S_2(x) = (1 - \rho)\hat{c}_2(x) + \rho\hat{c}_2(x)r(x).$$

Under the prior Unif$[0, M]$, lifetime success conditional on period-1 power x is

$$\begin{aligned} V(x) &= \frac{x}{M} + \frac{M - x}{M} S_2(x) \\ &= \frac{x}{M} + \frac{M - x}{M} \hat{c}_2(x)\big[1 - \rho + \rho r(x)\big] \\ &= \frac{x}{M} + \frac{h(x)}{M}\big[1 - \rho + \rho r(x)\big] \\ \implies \qquad MV(x) &= x + h(x)\big[1 - \rho + \rho r(x)\big] \end{aligned} \tag{A1}$$

Since

$$r'(x) = -\frac{1 - r(x)}{M - x},$$

differentiating (A1) and substituting that $h'(x) = -G(x)$:

$$MV'(x) = 1 - G(x) + \rho\big(1 - r(x)\big)\big(G(x) - \hat{c}_2\big). \tag{A2}$$

All terms on the right-hand side are weakly positive. Hence $V'(x) \geq 0$ almost everywhere, implying that V is weakly increasing.

We now turn to the convexity claim. Let

$$s(x) := 1 - r(x) = \frac{M - \bar{A}}{M - x}.$$

Then

$$s'(x) = \frac{s(x)}{M - x}, \qquad \hat{c}_2'(x) = -\frac{G(x) - \hat{c}_2(x)}{M - x} \quad \text{a.e.}$$

As x rises, fewer turnout realizations satisfy $A(\ell) > x$, so $G(x)$ is weakly decreasing. Because G may have jumps, it is not necessarily legal to generally differentiate it pointwise everywhere. Instead, we use its distributional derivative: the decline in G is represented by the nonnegative measure $-dG$. Taking the distributional derivative of (A2) gives

$$M\, dV'(x) = \big[1 - \rho s(x)\big](-dG) + \frac{2\rho s(x)}{M - x}\big(G(x) - \hat{c}_2(x)\big)\, dx. \tag{A3}$$

Both terms on the right-hand side are nonnegative. $-dG$ is nonnegative because G is weakly decreasing; moreover, reachable power satisfies $x \leq \bar{A}$, so $s(x) \in [0, 1]$ and $1 - \rho s(x) \geq 0$, while $G(x) - \hat{c}_2(x) \geq 0$ as shown above. Therefore the distributional second derivative of V is a nonnegative measure, which implies that V is weakly convex over the range of reachable first-period power. □

A.4 Benchmark results with tactical choice

Proposition A1 (A sufficient condition for peace on the entire path)**.** *Suppose peace is optimal following every possible period-1 failure. Let X_τ denote realized period-1 protest power when the organizer chooses tactic $\tau \in \{p, v\}$, under the equilibrium turnout distribution induced by that tactic.*

If furthermore for every z,

$$\mathbb{E}\left[(X_p - z)_+\right] \geq \mathbb{E}\left[(X_v - z)_+\right], \tag{A1.1}$$

then peace is optimal in period 1 as well.

Proof of Proposition A1. Let $V_p(x)$ denote the movement's probability of succeeding by the end of period 2 conditional on realized period-1 power x and peaceful continuation following failure. By Proposition 2, V_p is weakly increasing and convex. Condition (A1.1) requires the peaceful opening to generate at least as much expected power above every threshold z as the violent opening. For bounded random variables, this condition implies that every increasing convex function takes weakly higher expected value under X_p than under X_v.Hence, applying it to the increasing convex function V_p,

$$\mathbb{E}[V_p(X_p)] \geq \mathbb{E}[V_p(X_v)].$$

Because peace is optimal after every possible failure, these are exactly the lifetime success probabilities from opening peacefully and violently. Thus peace is optimal in period 1. □

Proof of Lemma 3. Suppose $A_v(\ell) \equiv \bar{v}$. By the assumptions on the technologies, $\bar{v} \in (0,1)$. If the organizer opens with violence and the protest fails, the posterior places probability 1 on $\theta \geq \bar{v}$. Flat violence can then never succeed, so the optimal continuation following a violent opening is peace.

Compare this $v \to p$ sequence with the feasible strategy $p \to v$. For every $\theta < \bar{v}$, both strategies eventually succeed: under $p \to v$, either peace succeeds immediately or the reserved violent tactic does so in period 2. For $\theta \geq \bar{v}$, violence cannot succeed under either sequence, so success depends entirely on the peaceful attempt. Under $p \to v$, that attempt occurs under the initial beliefs; under $v \to p$, it occurs only after failure at power $\bar{v}$, which strictly lowers the peaceful participation cutoff. By Lemma 2, using the same realization of the common cost shock therefore produces weakly lower turnout, and hence weakly lower peaceful power, when peace is postponed, with strict inequality with positive probability.

Since $v \to p$ is optimal conditional on opening with violence, whereas after opening peacefully the organizer can do weakly better than the restricted $p \to v$ strategy, it follows that the organizer strictly prefers using peace in period 1 rather than flat violence. □

B Linear setting

This appendix establishes the results for the linear specification $A_p(\ell) = \ell$ and $A_v(\ell) = \alpha + \beta\ell$, where $\alpha > 0$, $\beta \geq 0$, and $\alpha + \beta < 1$. The argument has three main parts:

First, I characterize the organizer's continuation choice after a failed first-period protest and identify the two continuation boundaries β_0 and β_α. Second, I establish the shape of the period-1 opening comparison: violence cannot be strictly optimal before it becomes repeatable, while after it becomes repeatable its opening advantage is strictly increasing in β. Third, these results reduce the proof of Proposition 4 to the opening comparison as β approaches the feasibility boundary $1 - \alpha$.

Some of the sign arguments needed for the first two parts are algebraically involved, but contribute little to the overall logic or intuition of the proof. I collect those calculations in Section B.4. Throughout the main argument, $\rho \in (0, 1)$, as implied by the model. Values $\rho = 0$ and $\rho = 1$ are used below only as continuous limiting cases.

B.1 Continuation choice (Proposition 3)

Following a failed first-period protest of realized power x, the posterior is Unif$[x, M]$. Let $S_p(x)$ and $S_v(x)$ denote the actual period-2 probabilities of success under peace and violence, respectively. It is useful to derive these objects explicitly, because the violent continuation problem changes form depending on whether the posterior lower bound x lies below or above violence's floor α.

Let F_x denote the posterior CDF and let $\hat{c}_2^\tau(x)$ denote the period-2 participation cutoff under tactic τ. From Lemma 1,

$$\hat{c}_2^\tau(x) = \int_0^1 F_x(A_\tau(\ell))\, d\ell.$$

Given a cutoff $\hat{c}_2^\tau(x)$, Lemma 2 implies that the actual success probability can be written as

$$S_\tau(x) = \rho(1 - \hat{c}_2^\tau(x))F_x(A_\tau(0)) + (1 - \rho)\hat{c}_2^\tau(x) + \rho\hat{c}_2^\tau(x)F_x(A_\tau(1)).$$

The three terms correspond, respectively, to zero turnout, an interior turnout realization, and full turnout. The middle term simplifies because, conditional on interior turnout, ℓ is uniform on $(0, 1)$ and the average success probability is exactly the participation cutoff $\hat{c}_2^\tau(x)$.

For peace, $A_p(\ell) = \ell$. Since a failed first-period protest has $x \geq 0$,

$$\hat{c}_2^p(x) = \int_x^1 \frac{\ell - x}{M - x}\, d\ell = \frac{(1 - x)^2}{2(M - x)}, \qquad F_x(A_p(0)) = 0, \qquad F_x(A_p(1)) = \frac{1 - x}{M - x}.$$

substituting yields

$$S_p(x) = \frac{(1 - x)^2}{2(M - x)} \left[(1 - \rho) + \rho\frac{1 - x}{M - x}\right]. \tag{A4}$$

For violence, there are three relevant cases.

Case 1: Suppose $x \leq \alpha$. Even zero turnout then produces power at least as large as the posterior lower bound, so

$$\hat{c}_2^v(x) = \int_0^1 \frac{\alpha + \beta\ell - x}{M - x}\, d\ell = \frac{\alpha + \beta/2 - x}{M - x}.$$

Using

$$F_x(A_v(0)) = \frac{\alpha - x}{M - x}, \qquad F_x(A_v(1)) = \frac{\alpha + \beta - x}{M - x},$$

and simplifying yields

$$S_v(x) = \frac{\alpha - x + (1-\rho)\beta/2}{M - x} + \frac{\rho\beta(\alpha + \beta/2 - x)}{(M - x)^2}, \qquad x \leq \alpha. \tag{A5}$$

Case 2: Suppose $\alpha < x < \alpha + \beta$. Then, low-turnout violent protests generate power below the posterior lower bound and therefore have no chance of success. Only turnout above $(x - \alpha)/\beta$ matters, so

$$\hat{c}_2^v(x) = \int_{(x-\alpha)/\beta}^1 \frac{\alpha + \beta\ell - x}{M - x}\, d\ell = \frac{(\alpha + \beta - x)^2}{2\beta(M - x)}.$$

Here $F_x(A_v(0)) = 0$ and $F_x(A_v(1)) = (\alpha + \beta - x)/(M - x)$, which gives

$$S_v(x) = \frac{(\alpha + \beta - x)^2}{2\beta(M - x)} \left[(1 - \rho) + \rho\frac{\alpha + \beta - x}{M - x}\right], \qquad \alpha < x < \alpha + \beta. \tag{A6}$$

Case 3: Finally, if $x \geq \alpha + \beta$, even full-turnout violence cannot defeat any regime type remaining in the posterior, so $S_v(x) = 0$.

Proof of Proposition 3. I show that the continuation advantage of violence is single-crossing in the power x of the failed first-period protest: as x rises, the organizer can switch at most once from violence to peace.

First, note that if $x > \alpha + \beta$, violence cannot succeed at any level of mobilization. For the remaining cases, I proceed in two steps, which are summarized below.

Case 1. In the intermediate region $x \in [\alpha, \alpha + \beta)$. I will show that $S_v(x)/S_p(x)$ – the relative advantage of violence over peace – is decreasing in x

Case 2. If $x \leq \alpha$, then the baseline power of violence is sufficient to defeat some regime types in the posterior. In this region, I will show that the baseline capacity α required for indifference is increasing in x.

The two regions then combine to give a single cutoff from violence to peace.

Case 1: Let $\beta > 0$ and consider $x \in [\alpha, \alpha + \beta)$.

Dividing (A6) by (A4) gives

$$\frac{S_v(x)}{S_p(x)} = \frac{1}{\beta}\left(\frac{\alpha+\beta-x}{1-x}\right)^2 \frac{(1-\rho)(M-x)+\rho(\alpha+\beta-x)}{(1-\rho)(M-x)+\rho(1-x)}. \tag{A7}$$

Both nonconstant ratios on the right-hand side are strictly decreasing in x. For the first, this follows from $\alpha+\beta<1$. For the second, write it as $(C-x)/(D-x)$, where $C=(1-\rho)M+\rho(\alpha+\beta)$ and $D=(1-\rho)M+\rho$, so $C<D$. Hence $S_v(x)/S_p(x)$ is strictly decreasing on this region.

Case 2: Consider $x \leq \alpha$.

In this case, the ratio S_v/S_p is less useful because violence's baseline capacity lies *inside* the posterior support. Instead, fix (x, β, M, ρ) and consider how large α must be to make the organizer indifferent. Let $\alpha^*(x)$ denote this indifference value. Section B.4.1 shows that $\alpha^*(x)$ is strictly increasing in x, while $S_v(x) - S_p(x)$ is strictly increasing in α. Thus, for any fixed actual α, violence is preferred exactly when $\alpha > \alpha^*(x)$, and this inequality can switch from true to false at most once as x rises.

The two comparisons fit together continuously at $x = \alpha$. In particular, both expressions for violent success give

$$S_v(\alpha) = \frac{\beta}{2(M-\alpha)}\left[(1-\rho)+\rho\frac{\beta}{M-\alpha}\right].$$

Moreover, at $x = \alpha+\beta < 1$, violence has zero probability of success, while peace still has positive probability of success, i..e $S_v(\alpha+\beta) = 0 < S_p(\alpha+\beta)$. Thus the comparison over the two regions combines into a single global cutoff $c_\rho(\alpha, \beta; M) \in [0, \alpha+\beta)$, such that violence is strictly preferred if and only if $x < c_\rho$.

Note that when $\beta = 0$, the region $(\alpha, \alpha+\beta)$ is empty. The same low-x argument applies for $x \leq \alpha$, while $S_v(x) = 0$ for $x \geq \alpha$, yielding the same cutoff characterization.

It remains to establish the comparative statics of the cutoff. Fix a posterior state x. Raising either α or β makes the violent technology pointwise weakly stronger. Its participation cutoff therefore weakly rises. Coupling the two environments using the same realization of the common participation-cost shock, the stronger technology induces weakly higher turnout and, because the technology itself is also stronger, weakly higher realized protest power. Thus $S_v(x)$ weakly increases with both α and β, while $S_p(x)$ is unchanged. The set of posterior states at which violence is optimal can therefore only expand, so $c_\rho(\alpha, \beta; M)$ is weakly increasing in both parameters. □

The two continuation boundaries. Having established the organizer's decision rule in the continuation, I proceed to the opening problem. Two cutoff values of β will be useful. The first determines whether violence is ever useful after a failure. By Proposition 3, it is enough to compare the tactics following the weakest possible failure, $x = 0$. From (A4)–(A5), the sign of $S_v(0) - S_p(0)$ is the sign of

$$B_0(\beta) := \rho\beta^2 + [M(1-\rho)+2\rho\alpha]\,\beta + 2M\alpha - [M(1-\rho)+\rho]. \tag{A8}$$

Thus $B_0(\beta) > 0$ exactly when violence is preferred following the weakest possible failure. Since $B_0'(\beta) = 2\rho\beta + M(1-\rho) + 2\rho\alpha > 0$, this comparison moves strictly in violence's favor as β rises.

Define

$$\alpha_0(M, \rho) := \frac{1-\rho}{2} + \frac{\rho}{2M}.$$

At $\beta = 0$, $B_0(0) = 2M(\alpha - \alpha_0(M, \rho))$. Hence, if $\alpha < \alpha_0(M, \rho)$, there is a unique $\beta_0(\alpha; M, \rho) \in (0, 1-\alpha)$ satisfying $B_0(\beta_0) = 0$. Existence follows from

$$B_0(1-\alpha) = \alpha\big(M(1+\rho) - \rho\alpha\big) > 0.$$

For $\beta < \beta_0$, peace is strictly preferred following every possible failure. At $\beta = \beta_0$, the organizer is indifferent only following $x = 0$, while for $\beta > \beta_0$, violence is optimal following sufficiently weak failures. If $\alpha = \alpha_0(M, \rho)$, the boundary is $\beta_0 = 0$; if $\alpha > \alpha_0(M, \rho)$, violence is already useful after sufficiently weak failures at $\beta = 0$, so there is no region in which the continuation tactic is always peace.

The second boundary determines whether violence is *repeatable*, that is, if violence is used in the first period protest (and failed), whether it can ever be followed again by violence.

The weakest violent protest generates power α, so violence is repeatable if $c_\rho > \alpha$. Let $\beta_\alpha(\alpha; M, \rho)$ denote the unique value satisfying $c_\rho = \alpha$. Substituting $x = \alpha$ into the continuation comparison gives

$$\beta\left[(1-\rho)(M-\alpha) + \rho\beta\right] = (1-\alpha)^2\left[(1-\rho)(M-\alpha) + \rho(1-\alpha)\right]. \tag{A9}$$

The left-hand side is strictly increasing in β and equals zero at $\beta = 0$. At $\beta = 1-\alpha$ it strictly exceeds the right-hand side. Thus there is a unique $\beta_\alpha(\alpha; M, \rho) \in (0, 1-\alpha)$.

Finally, note that whenever β_0 exists, $\beta_0 < \beta_\alpha$. At $\beta = \beta_\alpha$, violence and peace are equally attractive following some $x = \alpha > 0$. Single-crossing then implies that violence is strictly preferred following $x = 0$, so it must already have become useful as a fallback at a smaller value of β. Thus β_0 marks the first point at which violence is useful after any failure, while β_α marks the stronger condition under which violence can remain useful after one of its own failed openings.

B.2 The opening comparison

I now characterize period-1 tactical choice. Under the prior, first-period participation cutoffs are

$$\hat{c}_1^p = \frac{1}{2M}, \qquad \hat{c}_1^v = \frac{\alpha + \beta/2}{M}.$$

Let X_p and X_v denote realized period-1 protest power under peaceful and violent openings. Their CDFs on $[0, 1]$ are

$$F_p(x) = \rho(1 - \hat{c}_1^p) + (1-\rho)x, \qquad 0 \le x < 1, \tag{A10}$$

and

$$F_v(x) = \begin{cases} 0, & x < \alpha, \\ \rho(1 - \hat{c}_1^v) + (1-\rho)\dfrac{x-\alpha}{\beta}, & \alpha \le x < \alpha + \beta, \\ 1, & x \ge \alpha + \beta. \end{cases} \tag{A11}$$

For some level of realized first-period power x, define

$$H_\tau(x) := \frac{x}{M} + \frac{M-x}{M} S_\tau(x), \qquad \tau \in \{p, v\}.$$

Thus $H_\tau(x)$ is the movement's lifetime success probability conditional on first-period power x and continuation with tactic τ after failure.

Denote the value of the problem to the organizer by

$$V_\beta(x) := \max\{H_p(x), H_v(x)\}.$$

By Proposition 3,

$$V_\beta(x) = \begin{cases} H_v(x) & \text{if } x < c_\rho, \\ H_p(x) & \text{if } x \ge c_\rho, \end{cases}$$

Let $W_1^\tau := \mathbb{E}[V_\beta(X_\tau)]$. This describes (expected) lifetime success conditional upon opening with tactic τ. Finally, define violence's opening advantage by

$$\Delta_\rho(\alpha, \beta; M) := W_1^v - W_1^p.$$

The next lemma states the two facts about Δ_ρ needed for the opening result. If violence is not repeatable, then it also cannot be strictly optimal to use it first. If violence is repeatable, greater turnout sensitivity moves the opening comparison strictly toward violence.

Lemma A2 (Shape of the opening comparison)**.** *Let* $c := c_\rho(\alpha, \beta; M)$.

1. *If* $c \le \alpha$, *then* $\Delta_\rho(\alpha, \beta; M) \le 0$. *The inequality is strict whenever* $0 < c \le \alpha$.
2. *If* $c > \alpha$, *then*

$$\frac{\partial}{\partial \beta} \Delta_\rho(\alpha, \beta; M) > 0.$$

Proof. Proof of part 1. If $\beta = 0$, the result follows from Lemma 3, so suppose $\beta > 0$. If $c \le \alpha$, every violent opening produces $X_v \ge \alpha \ge c$ and is therefore followed by peace after failure. Hence $W_1^v = \mathbb{E}[H_p(X_v)]$. A peaceful opening, by contrast, is followed by violence after failures with $X_p < c$ and by peace otherwise.

First suppose $0 < c \le \alpha$. From (A5),

$$H_v(x) = \frac{\alpha + (1-\rho)\beta/2}{M} + \frac{\rho\beta(\alpha + \beta/2 - x)}{M(M-x)} \tag{A12}$$

and therefore

$$H_v'(x) = \frac{\rho\beta(2\alpha + \beta - 2M)}{2M(M-x)^2} < 0. \tag{A13}$$

To compare the two opening choices, separate histories according to whether first-period power falls below or above c. Below c, only a peaceful opening can generate such a weak failure, and the organizer responds by switching to violence. Above c, both opening choices lead to peaceful continuation. Since $H_v(c) = H_p(c)$, there is no discontinuity in value at the boundary. Integrating the two regions by parts gives

$$W_1^p - W_1^v = -\int_0^c F_p(x)H_v'(x)\,dx + \int_c^1 H_p'(x)\left[F_v(x) - F_p(x)\right]\,dx. \tag{A14}$$

The two terms correspond to these two regions. The first captures the value of the low-power histories that can only be generated by a peaceful opening; it is strictly positive because H_v is decreasing on $[0, c]$. The second compares the distributions of first-period power over histories for which both openings lead to the same peaceful continuation.

The first term is strictly positive by (A13). For the second term, (A10)–(A11) imply that $F_v - F_p$ crosses zero at most once, and only from negative to positive. Moreover, Proposition 2 implies that H_p' is nonnegative and increasing. Section B.4.2 verifies that period-2 indifference at c implies

$$\int_c^1 \left[F_v(x) - F_p(x)\right]\,dx > 0. \tag{A15}$$

Because the negative part of $F_v - F_p$ occurs below its positive part, weighting by the increasing function H_p' can only strengthen this comparison. Hence the second term in (A14) is nonnegative, and $W_1^p > W_1^v$.

Now suppose $c = 0$. Peace is then weakly optimal following every possible failure, so the relevant conditional lifetime-success function is H_p. Under the initial uniform prior, $S_\tau(0) = \mathbb{E}[X_\tau]/M$, and therefore

$$\int_0^1 \left[F_v(x) - F_p(x)\right]\,dx = \mathbb{E}[X_p] - \mathbb{E}[X_v] = M\left[S_p(0) - S_v(0)\right] \geq 0.$$

Because $F_v - F_p$ crosses at most once from negative to positive, this implies $\int_z^1 [F_v(x) - F_p(x)]\,dx \geq 0$ for every z, or equivalently

$$\mathbb{E}[(X_p - z)_+] \geq \mathbb{E}[(X_v - z)_+] \qquad \text{for every } z.$$

Proposition A1 therefore gives $W_1^p \geq W_1^v$, completing part (i).

Proof of part 2. Suppose $c > \alpha$ and let $\sigma := (c - \alpha)/\beta \in (0, 1)$. Thus σ is the interior turnout level under violence that generates exactly the continuation cutoff: $c = \alpha + \beta\sigma$. Violent openings with turnout below σ are followed by violence after failure; those with turnout above σ are followed by peace.

Using V_β, the two opening payoffs can be written as

$$W_1^p = \rho(1-\hat{c}_1^p)V_\beta(0) + (1-\rho)\int_0^1 V_\beta(x)\,dx + \rho\hat{c}_1^p V_\beta(1), \tag{A16}$$

$$W_1^v = \rho(1-\hat{c}_1^v)V_\beta(\alpha) + (1-\rho)\int_0^1 V_\beta(\alpha+\beta\ell)\,d\ell + \rho\hat{c}_1^v V_\beta(\alpha+\beta). \tag{A17}$$

When these expressions are differentiated with respect to β, changes in the continuation cutoff generate boundary terms. Those terms cancel because $H_v(c) = H_p(c)$: moving an infinitesimal set of histories across a point at which the organizer is indifferent has no first-order effect on the value.

The remaining calculation is given in Section B.4.3. It yields

$$\frac{\partial\Delta_\rho}{\partial\beta} = (1-\rho)^2 Q_0 + \rho(1-\rho)Q_1 + \rho^2 Q_2, \tag{A18}$$

where Q_0, Q_1, and Q_2 are all strictly positive whenever $c > \alpha$. Since the three weights in (A18) are nonnegative, the derivative is strictly positive. This proves part (ii). □

B.3 Proof of Proposition 4

Lemma A2 reduces the opening problem to a boundary comparison. At $\beta = \beta_\alpha$, we have $c_\rho = \alpha > 0$, so part (i) of the lemma implies

$$\Delta_\rho(\alpha, \beta_\alpha; M) < 0.$$

For every $\beta > \beta_\alpha$, violence is repeatable and part (ii) implies that Δ_ρ is strictly increasing in β. Thus there can be at most one value of β at which the organizer switches from a peaceful to a violent opening. It remains only to determine whether the opening advantage of violence becomes positive before reaching the feasibility boundary $\beta = 1-\alpha$.

Define the boundary comparison

$$\Phi(\alpha, M, \rho) := \lim_{\beta \nearrow 1-\alpha} \Delta_\rho(\alpha, \beta; M).$$

At $\beta = 1-\alpha$,

$$A_v(\ell) = \alpha + (1-\alpha)\ell \quad \geq \quad \ell = A_p(\ell),$$

with strict inequality whenever $\ell < 1$. Thus violence is point-wise at least as strong as peace. For any posterior following a failure, this also raises the citizens' participation cutoff, so under the same realization of the common participation-cost shock violence induces weakly higher turnout. Because $A_v(\ell) \geq A_p(\ell)$ at every turnout level, realized protest power is therefore weakly higher under violence as well. Hence violence is optimal in period 2 following every failure with power below 1.

At the boundary, violence is used after every failure. For a fixed continuation tactic, lifetime-success values are affine in ρ, while the period-1 turnout distribution is also affine in ρ. Hence $\Phi(\alpha, M, \rho)$ is a polynomial of degree at most two in ρ. Direct calculation yields

$$\Phi(\alpha, M, 0) = \frac{\alpha(1-\alpha)}{6M} > 0, \tag{A19}$$

while

$$\Phi(\alpha, M, 1) = -\frac{\alpha(1-\alpha)(M-1)(2M-\alpha-1)}{4M^3(M-\alpha)} < 0. \tag{A20}$$

It follows that Φ has a root in $(0, 1)$. Furthermore, because (A19)–(A20) have opposite signs, the root is unique.
Denote the root by

$$\bar{\rho}(\alpha, M) \in (0, 1).$$

Hence $\Phi(\alpha, M, \rho) > 0$ for $\rho < \bar{\rho}(\alpha, M)$ and $\Phi(\alpha, M, \rho) < 0$ for $\rho > \bar{\rho}(\alpha, M)$.

Suppose first that $\rho < \bar{\rho}(\alpha, M)$. The opening advantage is strictly negative at β_α and strictly positive at the feasibility boundary. Continuity and strict monotonicity therefore imply a unique

$$\beta^*_\rho(\alpha; M) \in (\beta_\alpha(\alpha; M, \rho), 1-\alpha)$$

such that $\Delta_\rho(\alpha, \beta^*_\rho; M) = 0$. Thus

$$W_1^v > W_1^p \quad \Longleftrightarrow \quad \beta > \beta^*_\rho(\alpha; M).$$

For $\beta \leq \beta_\alpha$, part (i) of Lemma A2 already rules out a strictly optimal violent opening. This establishes part (i) of Proposition 4.

Now suppose $\rho \geq \bar{\rho}(\alpha, M)$. If $\rho > \bar{\rho}$, the boundary value Φ is strictly negative; if $\rho = \bar{\rho}$, it equals zero. In either case, strict monotonicity on $(\beta_\alpha, 1-\alpha)$ implies

$$\Delta_\rho(\alpha, \beta; M) < 0 \qquad \forall \beta \in (\beta_\alpha, 1-\alpha).$$

Part (i) of Lemma A2 covers the remaining values $\beta \leq \beta_\alpha$. Peace is therefore optimal in period 1 for every admissible violent technology, establishing part (ii).

Additional implications. The same argument gives the ordering of the thresholds. Whenever a violence-first region exists,

$$\beta_\alpha < \beta^*_\rho < 1-\alpha.$$

Whenever β_0 exists in the feasible parameter space, Section B.1 additionally gives

$$\beta_0 < \beta_\alpha < \beta^*_\rho < 1-\alpha.$$

The violence-first threshold also approaches the feasibility boundary as mobilization responsiveness approaches its critical value:

$$\lim_{\rho \uparrow \bar{\rho}(\alpha, M)} \beta^*_\rho(\alpha; M) = 1-\alpha. \tag{A21}$$

To see this, fix any $\varepsilon \in (0, 1-\alpha)$. At $\rho = \bar{\rho}(\alpha, M)$, peace is strictly preferred at every interior $\beta < 1-\alpha$, so

$$\Delta_{\bar{\rho}}(\alpha, 1-\alpha-\varepsilon; M) < 0.$$

By continuity in ρ, the same inequality holds for all ρ sufficiently close to $\bar{\rho}$ from below. Since $\Delta_\rho(\alpha, \beta^*_\rho; M) = 0$, it follows that

$$\beta^*_\rho(\alpha; M) > 1-\alpha-\varepsilon$$

for all such ρ. Because $\beta^*_\rho(\alpha; M) < 1-\alpha$ always, and ε is arbitrary, (A21) follows.

B.4 Algebraic verifications

This section collects calculations used solely to establish a sign.

B.4.1 The low-x continuation comparison

I verify here the claim used in the proof of Proposition 3: for $x \leq \alpha$, the baseline violent capacity required for indifference is increasing in the power of the failed protest.

For fixed (x, β, M, ρ), equations (A4)–(A5) imply

$$\frac{\partial}{\partial \alpha}\left[S_v(x) - S_p(x)\right] = \frac{M + \rho\beta - x}{(M-x)^2} > 0.$$

There is therefore a unique value $\alpha^*(x)$ satisfying $S_v(x) = S_p(x)$, where $\alpha^*(x)$ need not itself lie in the admissible parameter region. Violence is preferred exactly when $\alpha > \alpha^*(x)$. It is therefore enough to show that $\alpha^*(x)$ is increasing in x.

Let

$$p := 1-x, \qquad k := M-1, \qquad u := \alpha^*(x) - x.$$

The indifference condition is equivalent to

$$2(k + p + \rho\beta)u = (1-\rho)(k+p)(p^2-\beta) + \rho(p^3-\beta^2). \tag{A22}$$

Since $p = 1-x$ and $u = \alpha^*(x) - x$, $dp/dx = -1$ and $du/dx = d\alpha^*/dx - 1$. Differentiating (A22) with respect to x, and then using the original indifference condition to eliminate u, gives

$$\frac{d\alpha^*(x)}{dx} = \frac{P(p, \beta, k, \rho)}{2(k + p + \rho\beta)^2}, \tag{A23}$$

where

$$\begin{aligned} P(p, \beta, k, \rho) = {} & \rho^2\beta(\beta + 2kp) \\ & + \rho\left[\beta\left\{2k(2-p) + p(4-3p)\right\} + kp(2k+p)\right] \\ & + 2(k+p)^2(1-p). \end{aligned} \tag{A24}$$

Here $k > 0$, $p > 0$, $\beta \geq 0$, and $\rho > 0$. Every displayed term is nonnegative, and the middle line is strictly positive. Hence $d\alpha^*(x)/dx > 0$, as required.

B.4.2 Upper-tail power before repeatability

I verify here equation (A15), the claim needed to sign the second term in (A14). Suppose $0 < c \leq \alpha$ and $S_v(c) = S_p(c)$. Direct integration of (A10)–(A11), followed by use of period-2 indifference at c, gives

$$\int_c^1 [F_v(x) - F_p(x)]\, dx = \frac{c}{2M}\left[1 + \rho + c^2 - \rho c - 2\alpha - (1+\rho)\beta\right]. \tag{A25}$$

It remains to show that the bracket is positive.

Let

$$z := 1 - c, \qquad k := M - 1, \qquad u := \alpha - c, \qquad s := z - \beta.$$

Because $c \leq \alpha$ and $\alpha + \beta < 1$, we have $u \geq 0$ and $s - u = 1 - \alpha - \beta > 0$. In these variables, the indifference condition is

$$\begin{aligned} 2(k + z + \rho\beta)u = k(1-\rho)\left[s - z(1-z)\right] - z^2(1-z) \\ + zs(1+\rho) - \rho s^2. \end{aligned} \tag{A26}$$

Using (A26), the bracket in (A25) can be rewritten as

$$\begin{aligned} &1 + \rho + c^2 - \rho c - 2\alpha - (1+\rho)\beta \\ &\quad = \frac{\rho}{k + z + \rho\beta}\Big\{2(k + z + \rho\beta)u + k\left[(1+\rho)s - \rho z(1-z)\right] + zs(1-z)\Big\}. \end{aligned} \tag{A27}$$

Thus it is enough to show $(1+\rho)s - \rho z(1-z) > 0$.

If $s \geq z(1-z)$, this follows immediately. If $s < z(1-z)$, the first term on the right-hand side of (A26) is nonpositive. Since the left-hand side is nonnegative,

$$-z^2(1-z) + zs(1+\rho) - \rho s^2 \geq 0.$$

Hence

$$z(1+\rho)s \geq z^2(1-z) + \rho s^2 > z^2(1-z),$$

where the strict inequality follows from $\rho > 0$ and $s > 0$. Dividing by $z > 0$ gives $(1+\rho)s > z(1-z) \geq \rho z(1-z)$. Thus the desired inequality holds in either case.

All three terms in braces in (A27) are therefore nonnegative, with the final two strictly positive. The bracket in (A25) is strictly positive, proving (A15).

B.4.3 Monotonicity of the opening advantage

I verify here the coefficient signs used in part (ii) of Lemma A2. Throughout this subsection, $c > \alpha$ and $\sigma = (c - \alpha)/\beta \in (0, 1)$.

For the calculation, it is useful to record the conditional lifetime-success functions explicitly. Peaceful continuation gives

$$H_p(x) = \frac{x}{M} + \frac{1-\rho}{2M}(1-x)^2 + \frac{\rho}{2M}\frac{(1-x)^3}{M-x}. \tag{A28}$$

Violent continuation is

$$H_v^L(x) = \frac{\alpha + (1-\rho)\beta/2}{M} + \frac{\rho\beta(\alpha+\beta/2-x)}{M(M-x)}, \qquad x \le \alpha, \tag{A29}$$

and

$$H_v^H(x) = \frac{x}{M} + \frac{1-\rho}{2\beta M}(\alpha+\beta-x)^2 + \frac{\rho}{2\beta M}\frac{(\alpha+\beta-x)^3}{M-x}, \qquad \alpha < x < \alpha+\beta. \tag{A30}$$

Differentiating the opening payoffs (A16)–(A17), splitting H_v at $x = \alpha$ and the optimal continuation value at $x = c$, gives boundary terms from the movement of c and σ. These cancel because $H_v(c) = H_p(c)$. After the cancellation and collection of terms in ρ, the derivative takes the form stated in (A18),

$$\frac{\partial \Delta_\rho}{\partial \beta} = (1-\rho)^2 Q_0 + \rho(1-\rho)Q_1 + \rho^2 Q_2.$$

The remainder of the calculation verifies that each coefficient is strictly positive.

Coefficient on $(1-\rho)^2$. Direct integration gives

$$Q_0 = \frac{2\beta + 3(1-\beta)\sigma - 3\alpha\sigma^2 + (1-\beta)\sigma^3}{6M}. \tag{A31}$$

Since $\alpha < 1-\beta$,

$$\begin{aligned} 6MQ_0 &> 2\beta + (1-\beta)\left(3\sigma - 3\sigma^2 + \sigma^3\right) \\ &= 2\beta + (1-\beta)\left[1-(1-\sigma)^3\right] > 0. \end{aligned}$$

Coefficient on $\rho(1-\rho)$. Let

$$e := 1-\alpha-\beta > 0, \qquad k := M-1 > 0, \qquad u := 1-\sigma \in (0,1).$$

The derivative of Q_1 with respect to σ simplifies to

$$Q_1'(\sigma) = \frac{kA(u) + B(u)}{2M(k+e+\beta u)^2}, \tag{A32}$$

where

$$\begin{aligned} A(u) &= 2\beta(1-\beta)u^3 + 6\beta e\, u(1-u) + 3e^2(1-u), \\ B(u) &= \beta^2(1-\beta)(u^4+u^3) + \beta e u^2\left[3\beta + u(2-5\beta)\right] \\ &\quad + 6\beta e^2 u(1-u) + 2e^3(1-u). \end{aligned}$$

All terms are nonnegative. The only factor whose sign is not immediate, $3\beta + u(2-5\beta)$, is linear in u and positive at both endpoints: $3\beta > 0$ and $3\beta + (2-5\beta) = 2(1-\beta) > 0$. Hence $Q_1'(\sigma) > 0$.

Evaluating the coefficient at $\sigma = 0$ and simplifying gives

$$Q_1(\sigma) > \frac{6M\alpha + 4M\beta + 3\alpha^2 + 12\alpha\beta - 12\alpha + 9\beta^2 - 6\beta + 6}{12M^2}. \tag{A33}$$

The numerator is increasing in M. Setting $M = 1$ and using $e = 1-\alpha-\beta$ reduces it to

$$3 + 10\beta + 3e^2 - 6\beta e = 3 + 3e^2 + \beta(10 - 6e) > 0.$$

Thus $Q_1 > 0$.

Coefficient on ρ^2. Let

$$d := 1 - \alpha - \beta > 0, \qquad D := M - \alpha - \beta > d.$$

Direct simplification gives

$$M^2 Q_2 = T(\beta, d, D),$$

where

$$T(\beta, d, D) = \frac{d^3}{4D} + \frac{\beta(D + \beta/4)}{D + \beta} - \frac{(1 - d - \beta/2)d^2(3D - d)}{2D^2} + \frac{1 - d}{2(D + 1 - d)}. \tag{A34}$$

Holding (d, D) fixed,

$$\frac{\partial T}{\partial \beta} = \frac{D^2 + D\beta/2 + \beta^2/4}{(D + \beta)^2} + \frac{d^2(3D - d)}{4D^2} > 0. \tag{A35}$$

It is therefore enough to show $T(0, d, D) > 0$. Let $\eta := d/D \in (0, 1)$. Then

$$T(0, d, D) = \frac{\eta}{4[d + \eta(1 - d)]} N(d, \eta), \tag{A36}$$

where

$$N(d, \eta) = 2d(1 - d)^2\eta^2 + 3d(1 - d)(3d - 2)\eta + 7d^3 - 6d^2 - 2d + 2. \tag{A37}$$

It remains only to verify that this quadratic is positive over $(d, \eta) \in (0, 1)^2$. Its vertex is

$$\eta^*(d) = \frac{3(2 - 3d)}{4(1 - d)}.$$

If $d \le 2/5$, then $\eta^*(d) \ge 1$, so the minimum over $\eta \in [0, 1]$ occurs at $\eta = 1$, where $N(d, 1) = 5d^2 - 6d + 2 > 0$. If $2/5 < d < 2/3$, the vertex is interior and

$$N(d, \eta^*) = -\frac{(5d - 4)(5d^2 - 8d + 4)}{8} > 0.$$

Finally, if $d \ge 2/3$, then $\eta^*(d) \le 0$, so the minimum occurs at $\eta = 0$. In this case let $g(d) := N(d, 0) = 7d^3 - 6d^2 - 2d + 2$. The only critical point of g in $[2/3, 1]$ is

$$d^* = \frac{6 + \sqrt{78}}{21},$$

where

$$g(d^*) = \frac{486 - 52\sqrt{78}}{441} > 0.$$

Thus $N(d, \eta) > 0$ throughout the relevant region, and (A35) implies $Q_2 > 0$.

All three coefficients in (A18) are strictly positive, completing the verification of part (ii) of Lemma A2.

C Proofs for repression extension

This section supplies proofs and supplementary results for the repression extension.

For completeness, the regime minimizes the movement's eventual probability of success plus the expected costs of repression and violent contention. Thus, for any policy profile π, write its expected loss as

$$J(\pi) = \Pr_{\pi}(\text{movement succeeds}) + E_{\pi}\left[\sum_{t} \kappa_{\tau_t}(1-\gamma_t^{\tau_t})\right] + \lambda\, \mathbb{E}_{\pi}[N_v],$$

where N_v is the number of periods in which violence is used. The results below first characterize how repression changes the movement's success probability; repression costs and λ determine which of these interventions the regime optimally chooses.

If tactic τ is repressed in period 2 by factor $\gamma \in [0,1]$, its participation cutoff is

$$\hat{c}^{\tau}_{2,\gamma}(x) = \int_0^1 F_x(\gamma A_\tau(\ell))\, d\ell.$$

Let $L(c,u)$ denote turnout generated by participation cutoff c and common participation-cost realization u. By Lemma 2, $L(c,u)$ is weakly increasing in c for every u. Define the movement's actual period-2 success probability under the repressed tactic by

$$S^{\tau}_{2,\gamma}(x) = \mathbb{E}_u\Big[F_x\Big(\gamma A_\tau\big(L(\hat{c}^{\tau}_{2,\gamma}(x),u)\big)\Big)\Big],$$

and write $S^{\tau}_2(x) := S^{\tau}_{2,1}(x)$.

C.1 Responsive repression

Lemma A3 (Success under tactic-specific repression)**.** *Fix x and τ. The function $S^{\tau}_{2,\gamma}(x)$ is continuous and weakly increasing in γ. It is strictly increasing over any interval on which it is positive.*

Proof. Take $\gamma_2 > \gamma_1$. Since F_x is weakly increasing,

$$\hat{c}^{\tau}_{2,\gamma_2}(x) \geq \hat{c}^{\tau}_{2,\gamma_1}(x).$$

Couple the two repression levels using the same realization u of the common participation-cost shock. Monotonicity of turnout in the participation cutoff gives

$$L\Big(\hat{c}^{\tau}_{2,\gamma_2}(x),u\Big) \geq L\Big(\hat{c}^{\tau}_{2,\gamma_1}(x),u\Big).$$

Because A_τ is weakly increasing,

$$\begin{aligned}\gamma_2 A_\tau\Big(L(\hat{c}^{\tau}_{2,\gamma_2}(x),u)\Big) &\geq \gamma_2 A_\tau\Big(L(\hat{c}^{\tau}_{2,\gamma_1}(x),u)\Big)\\ &\geq \gamma_1 A_\tau\Big(L(\hat{c}^{\tau}_{2,\gamma_1}(x),u)\Big).\end{aligned}$$

Applying the increasing function F_x and averaging over u establishes weak monotonicity. If the lower-repression-level success probability is positive, a positive-measure set of common shocks

generates power strictly above the posterior lower bound. On this set, increasing γ strictly increases realized power, and the posterior has positive density over the relevant interval, yielding strict monotonicity. Continuity follows from continuity of the cutoff and realized-power mappings together with dominated convergence. □

The next result is the formal version of the substitution cap stated verbally in Proposition 5.

Proposition A2 (Formal substitution cap)**.** *Fix a belief state x and tactics $\tau \neq \tau'$ satisfying*

$$S_2^{\tau}(x) > S_2^{\tau'}(x) > 0.$$

Suppose only tactic τ is repressed. Then there exists a unique threshold

$$\hat{\gamma}_{\tau}(x) \in (0,1)$$

such that the organizer prefers τ for $\gamma > \hat{\gamma}_{\tau}(x)$ and τ' for $\gamma < \hat{\gamma}_{\tau}(x)$. Defining

$$V_2(x,\gamma) = \max\left\{S_{2,\gamma}^{\tau}(x), S_2^{\tau'}(x)\right\},$$

we have

$$V_2(x,1) - V_2(x,\gamma) \leq S_2^{\tau}(x) - S_2^{\tau'}(x)$$

for every $\gamma \in [0,1]$, with equality for every $\gamma \leq \hat{\gamma}_{\tau}(x)$.

Proof. By Lemma A3, $S_{2,\gamma}^{\tau}(x)$ is continuous and strictly increasing wherever it takes the positive fallback value. At $\gamma = 1$,

$$S_{2,1}^{\tau}(x) > S_2^{\tau'}(x),$$

whereas at $\gamma = 0$,

$$S_{2,0}^{\tau}(x) = 0 < S_2^{\tau'}(x).$$

The intermediate value theorem therefore gives a unique $\hat{\gamma}_{\tau}(x)$ satisfying

$$S_{2,\hat{\gamma}_{\tau}(x)}^{\tau}(x) = S_2^{\tau'}(x).$$

Hence

$$V_2(x,\gamma) = \begin{cases} S_{2,\gamma}^{\tau}(x), & \gamma > \hat{\gamma}_{\tau}(x), \\ S_2^{\tau'}(x), & \gamma \leq \hat{\gamma}_{\tau}(x), \end{cases}$$

up to tie-breaking at equality, and the stated bound follows immediately. Proposition 5 is the verbal implication of this result. □

The violence-aversion parameter λ generates an additional asymmetry at the substitution boundary.

Corollary A1 (Regime incentives at the substitution boundary)**.** *Suppose repression costs are strictly increasing in intensity and the regime can target only one tactic in period 2.*

(i) If peace is initially preferred and violence is the viable fallback, no optimal crowd-control policy strictly overshoots the switching threshold to violence.

(ii) If violence is initially preferred and peace is the viable fallback, then for sufficiently large λ there exists a counterinsurgency policy that induces peace and strictly dominates no repression.

Proof. For part (i), let $\hat{\gamma}_p(x)$ be the switching threshold. For every $\gamma < \hat{\gamma}_p(x)$, the organizer has already abandoned peace, so its success probability is fixed at the unrepressed violent fallback value. Replacing γ by $\hat{\gamma}_p(x)$ weakly preserves that success probability while strictly lowering repression costs; depending on tie-breaking, it also weakly lowers the violence cost. Thus every strict overshoot is dominated.

For part (ii), choose any $\gamma < \hat{\gamma}_v(x)$ that induces peace. Relative to no repression, the induced-peace policy replaces the initially preferred violent success probability $S_2^v(x)$ with the lower fallback value $S_2^p(x)$, incurs finite repression cost $\kappa_v(1-\gamma)$, and avoids the direct violence cost λ. Hence it strictly dominates no repression whenever

$$\lambda > \kappa_v(1-\gamma) - \big(S_2^v(x) - S_2^p(x)\big).$$

The right-hand side is finite, so the claim follows. □

C.2 Preemptive repression and the belief refund

The main-text Proposition 6 suppresses two auxiliary objects. For a fixed continuation tactic τ, define its unconditional contribution to lifetime success after a first-period failure at power x by

$$C_\rho^\tau(x) := \frac{M-x}{M} S_2^\tau(x).$$

Let

$$C_\rho(x) := \max_{\tau \in \{p,v\}} C_\rho^\tau(x)$$

and define the movement's eventual success probability conditional on first-period realized power x by

$$H_\rho(x) = \frac{x}{M} + C_\rho(x).$$

Proposition A3 (Formal belief-refund decomposition)**.** *For every $x' < x$,*

$$C_\rho(x') \geq C_\rho(x),$$

and therefore

$$H_\rho(x) - H_\rho(x') = \frac{x-x'}{M} - \left[C_\rho(x') - C_\rho(x)\right].$$

Thus lowering realized first-period power weakly improves continuation prospects and offsets part of the direct loss in first-period success. Moreover, for

$$0 \leq x' < x \leq \min\{\alpha, c_\rho(\alpha,\beta;M)\},$$

we have

$$H_\rho(x') = H_\rho(x) \quad \text{if } \rho\beta = 0,$$

while

$$H_\rho(x') > H_\rho(x) \quad \text{if } \rho > 0 \text{ and } \beta > 0.$$

Proof. Fix a continuation tactic τ and consider $x' < x$. Under the lower posterior truncation point x', citizens are weakly more optimistic, so

$$\hat{c}_2^\tau(x') \geq \hat{c}_2^\tau(x).$$

Using the same common shock u in both continuation games gives

$$L(\hat{c}_2^\tau(x'), u) \geq L(\hat{c}_2^\tau(x), u),$$

and hence

$$A_\tau\big(L(\hat{c}_2^\tau(x'), u)\big) \geq A_\tau\big(L(\hat{c}_2^\tau(x), u)\big)\,.$$

Conditional on u, the continuation-success interval after failure at x is therefore contained in the corresponding interval after x'. Averaging over u yields

$$C_\rho^\tau(x') \geq C_\rho^\tau(x).$$

Taking the maximum over tactics preserves the inequality, establishing the decomposition.

For the full- and over-refund claims, let

$$z \leq \min\{\alpha, c_\rho(\alpha, \beta; M)\}.$$

Because $z \leq c_\rho$, violence is optimal in continuation. Because $z \leq \alpha$, even zero-turnout violence reaches the posterior lower bound. Its participation cutoff is therefore

$$\hat{c}_2^v(z) = \frac{\alpha + \beta/2 - z}{M - z}.$$

Using the turnout distribution from Lemma 2, the actual period-2 success probability is

$$S_2^v(z) = \rho\,(1 - \hat{c}_2^v(z))\,\frac{\alpha - z}{M - z} + (1 - \rho)\hat{c}_2^v(z) + \rho\hat{c}_2^v(z)\frac{\alpha + \beta - z}{M - z}.$$

Substituting into lifetime success gives

$$H_\rho(z) = \frac{\alpha + (1 - \rho)\beta/2}{M} + \frac{\rho\beta(\alpha + \beta/2 - z)}{M(M - z)}.$$

Differentiating,

$$H_\rho'(z) = -\frac{\rho\beta\,(M - \alpha - \beta/2)}{M(M - z)^2}.$$

Since $M > 1 > \alpha + \beta$, the final factor is strictly positive. Thus $H_\rho'(z) = 0$ when $\rho\beta = 0$ and $H_\rho'(z) < 0$ when $\rho > 0$ and $\beta > 0$, proving the result. Proposition 6 follows immediately. □

C.3 Persistence of violence

The no-period-2-repression assumption in Proposition 7 is a benchmark used to isolate the informational channel. Allowing the regime to re-optimize repression after the first failure would superimpose the responsive-substitution mechanism characterized above: in particular, second-period counterinsurgency could itself prevent the violent recurrence whose informational source we want to identify here.

Let γ scale only the period-1 violent technology. Under the prior, the period-1 participation cutoff when the organizer opens violently is

$$\hat{c}^v_{1,\gamma} = \int_0^1 \frac{\gamma(\alpha + \beta\ell)}{M}\, d\ell = \frac{\gamma(\alpha + \beta/2)}{M}.$$

Let X_γ denote realized first-period violent power under this policy, and let $c_\rho = c_\rho(\alpha, \beta; M)$ denote the continuation cutoff from Proposition 3.

Lemma A4 (Nested violent-recurrence events)**.** *Consider two counterinsurgency policies $\gamma' < \gamma$ under which the organizer opens violently in both cases. Coupling them using the same common participation-cost shock and regime type gives*

$$X_{\gamma'} < X_\gamma.$$

Consequently, if violence recurs in period 2 under γ, then it also recurs under the stronger policy γ'.

Proof. Because $\gamma' < \gamma$,

$$\hat{c}^v_{1,\gamma'} < \hat{c}^v_{1,\gamma}.$$

Under the same common participation-cost shock,

$$\ell_{\gamma'} \leq \ell_\gamma,$$

so

$$X_{\gamma'} = \gamma'(\alpha + \beta\ell_{\gamma'}) < \gamma(\alpha + \beta\ell_\gamma) = X_\gamma.$$

Violence recurs under γ exactly when

$$\theta \geq X_\gamma \qquad \text{and} \qquad X_\gamma < c_\rho.$$

These inequalities imply

$$\theta \geq X_\gamma > X_{\gamma'} \qquad \text{and} \qquad X_{\gamma'} < X_\gamma < c_\rho,$$

which are sufficient for recurrence under γ'. □

Proof of Proposition 7. By Lemma A4, the event of a second violent episode is weakly larger under stronger counterinsurgency as long as both policies preserve a violent opening. Since violence occurs with certainty in period 1 on both branches, the expected number of violent periods is therefore weakly larger under the stronger policy.

If counterinsurgency instead induces a peaceful opening, violence can occur only in period 2. Its expected incidence is strictly below one because peaceful protest has a strictly positive probability of succeeding in period 1. Every policy that preserves a violent opening, by contrast, generates one violent period with certainty and therefore has expected violent incidence at least one. Any policy inducing a peaceful opening consequently produces strictly less expected violence. □